ICS29.200
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National Standard of the People's Republic of China

GB/T27930-2015 replaces
GB/T27930-2011

Communication protocol between off-board conductive charger and battery management system of electric vehicles

Communicationprotocolsbetweenoff-
boardconductivechargerandbatterymanagementsystemforelectricvehicle

Released on December 28, 2015 Implementation on January 1, 2016

General Administration of Quality Supervision, Inspection and Quarantine
of the People's Republic of China Standardization Administration of China release

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Preface

This standard was drafted in accordance with the rules given in GB/T 1.1-2009.

This standard replaces GB/T 27930-2011 "Communication Protocol between Off-Board Conductive Chargers and Battery Management Systems for Electric Vehicles".

Compared to GB/T 27930-2011, in addition to editorial revisions, the main technical changes are as follows:

———It is stipulated that "chargers and BMSs implementing this standard should have forward compatibility" (see 4.6); ———In

case of poor communication environment, the communication rate shall be increased by 50kbit/s (see Chapter 5); ———

It is stipulated that "all bits of the optional items shall be sent in the format specified in this standard or filled with 1, and invalid bits or fields not specified in this standard shall be filled with

1"(see 7.9);

——— Modified the overall flow chart (see Chapter 8); ———

Added the communication handshake messages BHM and CHM (see 9.1); ———

Added 8 bytes for reservation in BRM (see 10.1.4); ——— Added the minimum

charging current field in CML (see 10.2.3); ——— Added the charge suspension

field in CCS (see 10.3.3); ——— Added the fault reason for BMS to

terminate charging (see 10.3.8); ——— Added the charging sequence flow chart

(see A.2); ——— Added the charging process fault handling

method (see Appendix C); ——— Added the message start and stop conditions

(see Appendix D). This standard is proposed and managed by the China Electricity Council. The

responsible drafting organizations of this standard are: State Grid

Corporation of China, China Energy Engineering Group Guangdong Electric Power Design Institute Co., Ltd., Nanjing Nan

Rui Group Corporation, China Automotive Technology and Research Center.

Participating drafting organizations of this standard: China Electricity Council, Xuji Group Co., Ltd., China Electric Power Research Institute, Shenzhen AUTO Power Equipment Co., Ltd., BYD Auto Industry Co., Ltd., BYD Daimler New Technology Co., Ltd., SAIC Motor Corporation Limited, Putian New Energy Co., Ltd., China Electric Power Research Institute, and Shanghai Electric Power Research Institute.

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The previous versions of the standards replaced by this standard are:

———GB/T 27930—2011.

Communication protocol between off-board conductive charger and battery management system of electric vehicles

1 Scope

This standard specifies the definitions of the communication physical layer, data link layer and application layer between the off-board conductive charger (hereinafter referred to as the charger) and the battery management system (hereinafter referred to as BMS) of electric vehicles based on the controller area network (Control Area Network, hereinafter referred to as CAN).

This standard applies to the communication between the charger and BMS using the charging mode 4 specified in GB/T18487.1, and also applies to the charging Communication between the machine and the vehicle control unit with charging control function.

2 Normative references

The following documents are essential for the application of this document. For dated references, only the dated versions apply to this document. For any undated referenced documents, the latest version (including all amendments) applies to this document.

GB/T19596 Electric Vehicle Terminology GB/

T18487.1 General Requirements for Conductive Charging Systems for Electric Vehicles

ISO 11898-1:2003 Road vehicle controller area networks Part 1: Data link layer and physical signalling vehicle—Controlareanetwork(CAN)Part1:Datalinklayerandphysicalsignaling]

SAE J1939-11:2006 Commercial Vehicle Control System Local Area Network CAN Communication Protocol Part 11: Physical Layer, 250K bit/s, Shielded Twisted Pair (Recommended practice for serial control and communication vehicle network—Part 11:Physicallayer-250Kbits/s,twistedshieldedpair)

SAEJ1939-21:2006 Commercial Vehicle Control System Local Area Network CAN Communication Protocol Part 21: Data Link Layer (Recom-mendedpracticeforserialcontrolandcommunicationvehiclenetwork—Part21:Datalinklayer)

SAEJ1939-73:2006 Commercial Vehicle Control System Local Area Network CAN Communication Protocol Part 73: Application Layer Diagnosis (Recom-mendedpracticeforserialcontrolandcommunicationvehiclenetwork—Part73:ApplicationLayer—Diagnostics

3 Terms and Definitions

The terms and definitions defined in GB/T 19596, SAE J 1939 and the following apply to this document.

3.1

Frame A series

of data bits that make up a complete message.

3.2

CAN data frame CANdataframe is an ordered

bit field required by the CAN protocol for transmitting data, starting with the start of frame (SOF) and ending with the end of frame (EOF).

3.3

Messages One or more

"CAN data frames" with the same parameter group number.

3.4

Identifier identifier part of the

CAN arbitration field.

3.5

Standard frame standardframe CAN

data frame using 11-bit identifier defined in the CAN2.0B specification.

3.6

Extended frame: A CAN data frame

using a 29-bit identifier defined in the CAN2.0B specification.

3.7

Priority is a 3-bit field in

the identifier that sets the arbitration priority of the transmission process, with the highest priority being level 0 and the lowest priority being level 7.

3.8

Parameter group parametergroup; PG is a

collection of parameters transmitted in a message.

3.9

Parameter group number parametergroupnumber; PGN is a 24-bit

value used to uniquely identify a parameter group. The parameter group number includes: reserved bits, data page, PDU format field (8 bits), PDU specific field (8 bits).

3.10

Suspect parameter number suspectparameternumber; SPN The

application layer assigns a 19-bit value to each parameter through parameter description signaling.

3.11

Protocol Data Unit protocoldataunit; PDU is a specific CAN

data frame format.

3.12

Transport protocol transportprotocol Part of

the data link layer, providing a mechanism for transmitting PGN data between 9 and 1785 bytes.

3.13

Electronic control unit electronic control unit; ECU electronic control

unit, that is, the on-board computer, consists of a microcontroller and peripheral circuits.

3.14

Diagnostic trouble code diagnostic trouble code; DTC A 4-byte

value used to identify the fault type, related fault mode and number of occurrences.

4 General Provisions

4.1 The communication network between the charger and the BMS utilizes the CAN 2.0B protocol. For the charging process, refer to Appendix A. 4.2 During charging, the charger and BMS monitor parameters such as voltage, current, and temperature, while the BMS manages the entire charging process. 4.3 The CAN communication network between the charger and the BMS consists of two nodes: the charger and the BMS.

4.4 Data transmission shall be in the format of sending the least significant byte

first. 4.5 Positive current values represent discharge, and negative current values

represent charge. 4.6 Chargers and BMSs implementing this standard shall be forward compatible.

5 Physical Layer

The physical layer of this standard must comply with the physical layer requirements of ISO 11898-1:2003 and SAE J1939-11:2006. Communication between the charger and the BMS in this standard must utilize a CAN interface independent of the powertrain control system. The communication rate between the charger and the BMS is 250 kbit/s.

Note: In special occasions with harsh communication environments (such as commercial vehicle charging stations with long communication distances), the power supply equipment manufacturer and the electric vehicle manufacturer may agree to

Therefore, a communication rate of 50 kbit/s can be used.

6 Data Link Layer

6.1 Frame Format

Devices adopting this standard shall use the 29-bit identifier of the CAN extended frame. The specific definition of each bit allocation shall comply with the relevant provisions in SAE J1939-21:2006.

6.2 Protocol Data Unit (PDU)

Each CAN data frame contains a single protocol data unit (PDU), see Table 1. The protocol data unit consists of seven parts,

They are priority, reserved bits, data page, PDU format, PDU specific, source address and data field.

Table 1 Protocol Data Unit (PDU)

	R DP					…
P			PF	PS	SA	DATA
3	1	1	8	8	8	0~64

Data format

requirements: 1. P is priority: set from the highest 0 to the lowest 7.

2.R is a reserved bit: for future development and use, this standard sets it to 0.

3.DP stands for Data Page: an auxiliary page used to select parameter group descriptions. This standard sets it to 0.

4. PF stands for PDU format: it is used to determine the format of the PDU and the parameter group number corresponding to the data field.

5. PS is a PDU-specific format: The PS value depends on the PDU format. In this standard, the PDU1 format is used, and the PS value is the destination address.

6.SA is the source address: the source address of the message.

7. DATA is the data field: If the given parameter group data length is ÿ 8 bytes, it is transmitted according to the message length specified in Chapter 9. If the given parameter group data length is 9 to 1785 bytes, multiple CAN data frames are required for data transmission. The connection management capability of the transmission protocol function is used to establish and close the communication of multiple parameter groups. For details, see the provisions of 6.5.

8. The third row of this table shows the number of digits.

6.3 Protocol Data Unit (PDU) Format

The PDU1 format defined in SAEJ1939-21:2006 is selected.

6.4 Parameter Group Number (PGN)

The second byte of PGN is the PDU format (PF) value, with both the high byte and the low byte being 00H.

6.5 Transport Protocol Function

The transmission protocol function is used to transmit data between the BMS and the charger, ranging from 9 to 1785 bytes. Connection initialization, data transmission, and connection termination must comply with the message transmission requirements in 5.4.7 and 5.10 of SAE J1939-21:2006. For multi-frame messages, the message period is the transmission period of the entire data packet.

6.6 Address Allocation

Network addresses are used to ensure the uniqueness of information identifiers and indicate the source of information. Chargers and BMSs are defined as non-configurable addresses. This means that these addresses are fixed in the ECU's program code and cannot be altered by any means, including service tools. The addresses assigned to chargers and BMSs are shown in Table 2.

Table 2 Charger and BMS address allocation

Device	Preferred address
charger	86(56H)
BMS	244(F4H)

6.7 Information Type

The CAN bus technical specification supports five types of information, namely command, request, broadcast/response, confirmation and group function. The provisions of 5.4 Information Type in SAEJ1939-21:2006 shall be followed.

7 Application Layer

7.1 The application layer defines parameters and parameter groups. 7.2 Parameter groups are numbered using PGNs, and each node identifies the contents of a data packet based on the PGN. 7.3 "Request PGN" is used to proactively obtain parameter groups from other nodes. 7.4 Data is transmitted using both periodic and event-driven methods. 7.5 If multiple PGNs are required to implement a function, the function is considered successfully transmitted only when multiple PGNs are received simultaneously. 7.6 When defining a new parameter group, parameters with the same function, the same or similar refresh rates, and parameters belonging to the same subsystem should be grouped together. Furthermore, new parameter groups should fully utilize the 8-byte data width and group related parameters together as much as possible. At the same time, scalability should be considered, and some bytes or bits should be reserved for future modifications. 7.7 When modifying a parameter group defined in Chapter 9, the definitions of existing bytes or bits should not be modified. Newly added parameters must be related to existing parameters in the parameter group. Unrelated parameters should not be added to an existing PGN to save PGNs. 7.8 The definitions of various fault diagnostics for the charger and BMS during charging should comply with the requirements for the CAN bus diagnostic system in 5.1 of SAE J1939-73:2006. Appendix B provides the fault diagnostic message definition specifications. 7.9 Message options are divided into mandatory and optional items. If all content in a message frame is optional, the message may not be sent. If part of the content in a message frame is optional, all bits of the optional items shall be sent or padded with 1 according to the format specified in this standard. Invalid bits or fields not specified in this standard shall be padded with 1. Bits not specified in this standard or reserved bits shall be padded with 1. 7.10 The length, mandatory content, and format of the message shall be sent in accordance with the specifications in Chapter 10.

8 Overall Charging Process

The entire charging process includes six stages: physical connection completion, low-voltage auxiliary power-up, charging handshake stage, charging parameter configuration stage, charging In each stage, if the charger and BMS do not receive each other's message within the specified time or do not When a correct message is received, it is judged as timeout (timeout means that the complete data packet or correct data packet of the other party is not received within the specified time). Unless otherwise specified, the time is 5s. When a timeout occurs, the BMS or charger sends the error message specified in 9.5 and enters the error handling state. In the process of troubleshooting, different treatments are performed according to the type of fault (see Appendix C). If a fault occurs, the charging process will be terminated directly. For the start and stop conditions of the message, please refer to Appendix D. The process is shown in Figure 1.

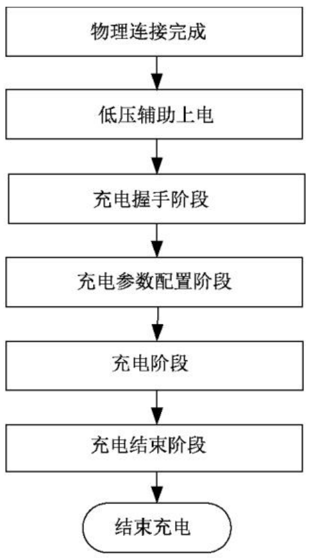

Figure 1 Overall charging flow chart

9 Message Classification

9.1 Low-voltage auxiliary power-up and charging handshake stage

The charging handshake phase is divided into the handshake startup phase and the handshake identification phase. When the charger and BMS are physically connected and powered on, the charging The low voltage auxiliary power supply enters the handshake start phase and sends a handshake message, and then performs insulation monitoring. After the insulation monitoring is completed, the handshake identification phase is entered. In the first stage, both parties send identification messages to confirm the necessary information of the battery and charger. CHM message and BHM message are new additions for product compatibility. Message, used by the charger and BMS to determine the standard version used by both parties during the handshake startup phase. For typical charging working state transitions, see For details on the charging sequence in Figures A.1 and A.2, see GB/T 18487.1. The message during the charging handshake phase should comply with the requirements in Table 3.

Table 3 Message classification during the charging handshake phase

Message code name	Message Description	PGN (Dec)	PGN (Hex)	priority	Data length byte	Message cycle ms	Source address-destination address
CHM	Charger handshake	9728	002600H	6	3	250	Charger-BMS
BHM	Vehicle handshake	9984	002700H	6	2	250	BMS-Charger

Table 3 (continued)

Message code name	Message Description	PGN (Dec)	PGN (Hex)	priority	Data length byte	Message cycle ms	Source address-destination address
CRM	Charger identification	256	000100H	6	8	250	Charger-BMS
BRM	BMS and vehicle identification message	512	000200H	7	41	250	BMS-Charger

9.2 Charging parameter configuration stage

After the charging handshake phase is completed, the charger and BMS enter the charging parameter configuration phase. In this phase, the charger sends a charging parameter configuration message to the BMS. The maximum output capacity of the motor is reported, and the BMS determines whether charging is possible based on the maximum output capacity of the charger. See Figure A.3 for state transition. The message in the charging parameter configuration phase should comply with the requirements of Table 4.

Table 4 Message classification during the charging parameter configuration phase

Message code name	Message Description	PGN (Dec)	PGN (Hex)	priority	Data length byte	Message cycle ms	Source address-destination address
BCP	power battery charging parameters	1536	000600H	7	13	500	BMS-Charger
CTS	charger sends time synchronization information	1792	000700H	6	7	500	Charger-BMS
CML	charger maximum output capacity	2048	000800H	6	8	250	Charger-BMS
BRO	battery charging ready status	2304	000900H	4	1	250	BMS-Charger
CRO	charger output ready status	2560	000A00H	4	1	250	Charger-BMS

9.3 Charging Stage

After the charging configuration phase is completed, the charger and BMS enter the charging phase. During the entire charging phase, the BMS sends power to the charger in real time. The charger adjusts the charging voltage and current according to the battery charging requirements to ensure the normal charging process. During the process, the charger and BMS send each other their respective charging status. In addition, BMS sends power battery status to the charger as required. Specific status information and voltage, temperature, etc. BMV, BMT, and BSP are optional reports, and the charger does not determine message timeouts for them.

BMS determines whether the charging process is normal, whether the battery status reaches the charging end condition set by BMS itself, and whether the charging The motor stops charging message (including the specific reason for the stop, the message parameter value is all 0 and the untrusted state) to determine whether to end charging; the charger Based on whether the stop charging instruction is received, whether the charging process is normal, whether the charging parameter value set by the human is reached, or whether the BMS is received The charging termination message (including the specific termination reason, message parameter values are all 0 and untrusted status) is used to determine whether to end charging. Refer to Figure A.4 for the transition of electrical working states. The message in the charging stage shall comply with the requirements of Table 5.

Table 5: Message classification during charging phase

Message code name	Message Description	PGN (Dec)	PGN (Hex)	priority	Data Bytes byte	Message cycle	source address-destination address
BCL	Battery charging requirements	4096	001000H	6	5	50ms	BMS-Charger
BCS	Overall battery charge status	4352	001100H	7	9	250ms	BMS-Charger
CCS	Charger charging status	4608	001200H	6	8	50ms	Charger-BMS
BSM	power battery status information	4864	001300H	6	7	250ms	BMS-charger

Table 5 (continued)

Message code name	Message Description	PGN (Dec)	PGN (Hex)	priority	Data Bytes byte	Message cycle	source address-destination address
BMV	single power battery voltage	5376	001500H	7	Uncertain	10s	BMS-Charger
BMT	power battery temperature	5632	001600H	7	Uncertain	10s	BMS-Charger
BSP	power battery reserved message	5888	001700H	7	Uncertain	10s	BMS-Charger
BST	BMS stops charging	6400	001900H	4	4	10ms	BMS-Charger
CST	The charger stops charging	6656	001A00H	4	4	10ms	Charger-BMS

9.4 Charging End Stage

When the charger and BMS stop charging, both parties enter the charging end stage. In this stage, the BMS sends the entire charging process to the charger. Charging statistics during the process, including: initial SOC, final SOC, minimum and maximum battery voltage; the charger receives charging data from BMS After collecting statistics, the output power, cumulative charging time and other information of the entire charging process are sent to the BMS, and finally the low-voltage auxiliary power supply is stopped. The typical charging state transition is shown in Figure A.5. The message at the end of charging should comply with the requirements of Table 6.

Table 6 Classification of messages at the end of charging

Message code name	Message Description	PGN (Dec)	PGN (Hex)	priority	Data Bytes byte	Message cycle ms	Source address-destination address
BSD	BMS Statistics	7168	001C00H	6	7	250	BMS-Charger
CSD	Charger statistics	7424	001D00H	6	8	250	Charger-BMS

9.5 Error Messages

During the entire charging phase, when the BMS or charger detects an error, it sends an error message. The error message should comply with Table 7. Require.

Table 7 Error message classification

Message code name	Message Description	PGN (Dec)	PGN (Hex)	priority	Data Bytes byte	Message cycle ms	Source address-destination address
BEM	BMS error message	7680	001E00H	2	4	250	BMS-Charger
CEM	Charger error message	7936	001F00H	2	4	250	Charger-BMS

10 Message Format and Content

10.1 Low-voltage auxiliary power-up and charging handshake phase message

10.1.1 PGN9728 Charger Handshake Message (CHM)

Message function: When the charger and electric vehicle are physically connected and powered on, and the voltage detection is normal, the charger sends a message to the BMS every The charger sends a handshake message every 250ms to confirm whether the handshake between the two parties is normal. The PGN9728 message format is shown in Table 8.

Table 8 PGN9728 message format

Start byte or bit	length	SPN	SPN Definition	Sending Options
1	3 bytes	2600	Charger communication protocol version number, this standard specifies the current version This is V1.1, expressed as: byte3, byte2—0001H; byte1—01H	Required

10.1.2 PGN9984BMS handshake message (BHM)

Message function: After receiving the handshake message from the PGN9728 charger, the BMS returns the handshake message to the charger every 250ms. Provides the maximum allowable total charging voltage for the BMS. See Table 9 for the PGN9984 message format.

Table 9 PGN9984 message format

Start byte or bit	length	SPN	SPN Definition	Sending Options
1	2 bytes	2601	Maximum allowable total charging voltage	Required

in:

SPN2601 maximum output voltage (V):

Data resolution: 0.1V/bit, 0V offset.

10.1.3 PGN256 Charger Identification Message (CRM)

Message function: When the charger passes the handshake confirmation and confirms that the insulation test is normal, the charger sends a message to the BMS every 250ms. Identification message, used to confirm that the communication link between the charger and BMS is correct. Before receiving the BMS identification message, the confirmation code = 0x00; after receiving After the BMS recognizes the message, the confirmation code is 0xAA. The PGN256 message format is shown in Table 10.

Table 10 PGN256 message format

Starting byte or bit	length	SPN	SPN Definition	Sending Options
1	1 byte	2560	Identification result, (cannot identify; ÿ:= BMS can identify)	Required
2	4 bytes	2561	Charger number, 1/bit, 0 offset, data range: 0 ~0xFFFFFFFF	Required
6	3 bytes	2562	Charging station/charger area code, standard ASCII code	Optional

10.1.4 PGN512BMS and Vehicle Recognition Message (BRM)

Message function: Provide BMS and vehicle identification information to the charger during the charging handshake phase. After the motor identifies the message, it is sent to the charger every 250ms. When the data field length exceeds 8 bytes, the transmission protocol function must be used for transmission. The format is detailed in 6.5. The interval between frames is 10ms. The charger identification code with SPN2560=0xAA is received within 5 seconds. The PGN512 message format is shown in Table 11.

Table 11 PGN512 message format

Starting byte or bit length		SPN	SPN Definition	Sending Options
1	3 bytes	2565	BMS communication protocol version number, this standard stipulates that the current version is V1.1, Shown as: byte3, byte2—0001H; byte1—01H	Required
4	1 byte	2566	Battery type, 01H: lead-acid battery; 02H: nickel-metal hydride battery; 03H: iron phosphate battery Lithium battery; 04H: lithium manganese oxide battery; 05H: lithium cobalt oxide battery; 06H: ternary Material battery;07H: polymer lithium ion battery;08H: lithium titanate battery; FFH: Other batteries	Required
5	2 bytes	2567	Rated capacity of vehicle power battery system/Ah, 0.1Ah/bit, 0Ah bias Displacement	Required
7	2 bytes	2568	Rated total voltage of vehicle power battery system/V, 0.1V/bit, 0V offset Displacement	Required
9	4 bytes	2569	Battery manufacturer name, standard ASCII code	Optional
13	4 bytes	2570	Battery pack serial number, reserved, defined by the manufacturer	Optional
17	1 byte	2571	Battery pack production date: year, 1 year/bit, 1985 offset, data range Period: 1985~2235	Optional
18	1 byte		Battery pack production date: month, 1 month/bit, 0 month offset, data range: January to December	Optional
19	1 byte		Battery pack production date: day, 1 day/bit, 0 day offset, data range: 1~31st	Optional
20	3 bytes	2572	Battery pack charge times, 1 time/bit, 0 times offset, based on BMS statistics subject to	Optional
twenty three	1 byte	2573	Battery pack ownership identifier (<0>:=lease; <1>:=vehicle owned)	Optional
twenty four	1 byte	2574	Reserved	Optional
25	17 bytes	2575	Vehicle Identification Number (VIN)	Optional
42	8 bytes	2576	The 8-byte BMS software version number indicates the current BMS version information. Determined by hexadecimal encoding. Byte8, byte7, byte6—000001H ~ FFFFFEH, reserved, fill FFFFFFH; Byte5-byte2 is used as the BMS software version compilation time information mark, Byte5, byte4—0001H~FFFEH represents “year” (e.g. 2015 Year: fill in Byte5—DFH, byte4—07H); Byte3—01H~0CH represents “month” (e.g. November: fill in Byte3—0BH); Byte2—01H~1FH represents “day” (e.g. 10th: fill in Byte2—0AH); Byte1—01H~FEH indicates the version serial number (for example, 16: fill in Byte1—10H). (The above values indicate that BMS is currently using the 16th date on November 10, 2015 (This compiled version does not have the authentication authorization code filled in)	Optional

10.2 Parameter Configuration Phase Message

10.2.1 PGN1536 Battery Charging Parameters Message (BCP)

Message function: During the charging parameter configuration phase, the BMS sends the power battery charging parameters to the charger. If this message is not received, it is a timeout error and the charger should immediately stop charging. The PGN1536 message format is shown in Table 12.

Table 12 PGN1536 message format

Starting byte or bit length		SPN	SPN Definition	Sending Options
1	2 bytes	2816	Maximum allowable charging voltage of single power battery	Required
3	2 bytes	2817	Maximum allowable charging current	Required
5	2 bytes	2818	Nominal total energy of power battery	Required
7	2 bytes	2819	Maximum allowable total charging voltage	Required
9	1 byte	2820	Maximum allowable temperature	Required
10	2 bytes	2821	Vehicle power battery state of charge	Required
12	2 bytes	2822	Current battery voltage of the vehicle's power battery	Required

in:

1) SPN2816 Maximum allowable charging voltage of single power battery

Data resolution: 0.01V/bit, 0V offset; Data range: 0~24V;

2) SPN2817 maximum allowable charging current

Data resolution: 0.1A/bit, -400A offset;

3) SPN2818 power battery nominal total energy

Data resolution: 0.1kW·h/bit, 0kW·h offset; Data range: 0~1000kW·h;

4) SPN2819 Maximum allowable total charging voltage

Data resolution: 0.1V/bit, 0V offset;

5) SPN2820 Maximum allowable power battery temperature

Data resolution: 1 ÿ/bit, -50 ÿ offset; Data range: -50 ÿ ~+200 ÿ;

6) SPN2821 Vehicle power battery state of charge (SOC)

Data resolution: 0.1%/bit, 0% offset; Data range: 0~100%;

7) SPN2822 Current battery voltage of vehicle power battery

Data resolution: 0.1V/bit, 0V offset.

10.2.2 PGN1792 Charger Sends Time Synchronization Information Message (CTS)

Message function: Time synchronization information sent by the charger to the BMS during the charging parameter configuration phase. The PGN1792 message format is shown in Table 13.

Table 13 PGN1792 message format

Starting byte or bit length		SPN	SPN Definition	Sending Options
1	7 bytes	2823	Year/Month/Day/Hour/Minute/Second	Optional

Among them, SPN2823 date/time

1st byte: second (compressed BCD code); 2nd byte: minute (compressed BCD code);

3rd byte: hour (compressed BCD code); 4th byte: day (compressed BCD code);

Byte 5: Month (compressed BCD code); Bytes 6-7: Year (compressed BCD code).

10.2.3 PGN2048 Charger Maximum Output Capacity Message (CML)

Message function: The charger sends the maximum output capacity of the charger to the BMS in order to estimate the remaining charging time. PGN2048 message format See Table 14.

Table 14 PGN2048 message format

Starting byte or bit length		SPN	SPN Definition	Sending Options
1	2 bytes	2824	Maximum output voltage (V)	Required
3	2 bytes	2825	Minimum output voltage (V)	Required
5	2 bytes	2826	Maximum output current (A)	Required
7	2 bytes	2827	Minimum output current (A)	Required

in:

1) SPN2824 maximum output voltage (V)

Data resolution: 0.1V/bit, 0V offset;

2) SPN2825 minimum output voltage (V)

Data resolution: 0.1V/bit, 0V offset;

3) SPN2826 maximum output current (A)

Data resolution: 0.1A/bit, -400A offset;

4) SPN2827 minimum output current (A)

Data resolution: 0.1A/bit, -400A offset.

10.2.4 PGN2304BMS Charge Ready Message (BRO)

Message function: BMS sends a message to the charger indicating that the battery is ready to charge, so that the charger can confirm that BMS is ready to charge. If the charger is not ready within 60 seconds, it will wait; otherwise, refer to C.1 for processing. The PGN2304 message format is shown in Table 15.

Table 15 PGN2304 message format

Starting byte or bit length		SPN	SPN Definition	Sending Options
1	1 byte	2829	Is BMS ready for charging? (= BMS is not ready for charging Prepare); ÿ:=BMS completes charging preparation; ÿ: =Invalid)	Required

10.2.5 PGN2560 Charger Output Ready Message (CRO)

Message function: The charger sends a message to the BMS indicating that the charger is ready to output. If the motor is not ready within 60 seconds, the BMS will wait; otherwise, refer to Appendix C.1 for processing. Table 16.

Table 16 PGN2560 message format

Starting byte or bit length		SPN	SPN Definition	Sending Options
1	1 byte	2830	Is the charger ready to charge? (= charger is not ready to charge Power ready; < 0xAA >:= charger is ready for charging; ÿ:=Invalid)	Required

10.3 Charging Phase Message

10.3.1 PGN4096 Battery Charging Request Message (BCL)

Message function: allows the charger to adjust the charging voltage and charging current according to the battery charging requirements to ensure the normal charging process. If the charger does not receive the message within 1 second, it is a timeout error and the charger should end charging immediately.

In constant voltage charging mode, the output voltage of the charger should meet the voltage requirement value, and the output current should not exceed the current requirement value; In constant current charging mode, the output current of the charger should meet the current requirement value, and the output voltage should not exceed the voltage requirement value. When the charging current request in the text is greater than the maximum output current in the CML message, the charger outputs according to the maximum output capacity; when the charging current in the BCL message is greater than the maximum output current in the CML message, the charger outputs according to the maximum output capacity; When the current request is less than or equal to the maximum output current in the CML message, the charger outputs the requested current; when the voltage demand or current demand is When the value is 0, the charger outputs at the minimum output capacity. The PGN4096 message format is shown in Table 17.

Table 17 PGN4096 message format

Starting byte or bit length		SPN	SPN Definition	Sending Options
1	2 bytes	3072	Voltage requirement (V)	Required
3	2 bytes	3073	Current demand (A)	Required
5	1 byte	3074	Charging mode (0x01: constant voltage charging; 0x02: constant current charging)	Required

in:

1) SPN3072 voltage requirements

Data resolution: 0.1V/bit, 0V offset;

2) SPN3073 current requirements

Data resolution: 0.1A/bit, -400A offset.

10.3.2 PGN4352 Battery Charging Status (BCS)

Message function: Let the charger monitor the charging status of the battery pack such as charging voltage and charging current during charging. If the charger is within 5s If this message is not received, it is a timeout error and the charger should immediately terminate charging. The PGN4352 message format is shown in Table 18.

Table 18 PGN4352 message format

Starting byte or bit length		SPN	SPN Definition	Sending Options
1	2 bytes	3075	Charging voltage measurement value (V)	Required
3	2 bytes	3076	Charging current measurement value (A)	Required
5	2 bytes	3077	Maximum single power battery voltage and group number	Required
7	1 byte	3078	Current state of charge SOC (%)	Required
8	2 bytes	3079	Estimated remaining charging time (min)	Required

in:

1) SPN3075 charging voltage measurement value

Data resolution: 0.1V/bit, 0V offset;

2) SPN3076 charging current measurement value

Data resolution: 0.1A/bit, -400A offset;

3) SPN3077 Maximum single power battery voltage and group number

1-12 bits: Maximum single power battery voltage, data resolution: 0.01V/bit, 0V offset; data range: 0~24V;

13-16 bits: group number of the highest single power battery voltage, data resolution: 1/bit, 0 offset; data range: 0~15;

4) SPN3078 Current state of charge SOC

Data resolution: 1%/bit, 0% offset; Data range: 0~100%;

5) SPN3079 estimates the remaining charging time. When the remaining time calculated by the BMS based on the actual current exceeds 600 minutes, Send according to 600min.

Data resolution: 1min/bit, 0min offset; data range: 0~600min.

10.3.3 PGN4608 Charger Charging Status Message (CCS)

Message function: Let BMS monitor the current output of the charger, such as charging current, voltage value, etc. If BMS does not receive the message within 1s, This message indicates a timeout error and the BMS should immediately terminate charging. The PGN4608 message format is shown in Table 19.

Table 19 PGN4608 message format

Starting byte or bit length		SPN	SPN Definition	Sending Options
1	2 bytes	3081 Voltage output value (V)		Required
3	2 bytes	3082 Current output value (A)		Required
5	2 bytes	3083 Accumulated charging time (min)		Required
7.1	2 people	3929 Charging allowed (<00>:=pause; <01>:=allow)		Required

Note: When SPN3929 in CCS is 0, it means the charger will stop outputting. When SPN3929 is 1, it means the charger will continue charging.

in:

1) SPN3081 voltage output value (V)

Data resolution: 0.1V/bit, 0V offset;

2) SPN3082 current output value (A)

Data resolution: 0.1A/bit, -400A offset;

3) SPN3083 cumulative charging time (min)

Data resolution: 1min/bit, 0min offset; data range: 0~600min.

10.3.4 PGN4864BMS sends power battery status information message (BSM)

Message function: The BMS sends power battery status information to the charger during the charging phase. The PGN4864 message format is shown in Table 20.

Table 20 PGN4864 message format

Starting byte or bit	length	SPN	SPN Definition	Sending Options
1	1 byte	3085	The highest single power battery voltage number	Required
2	1 byte	3086	Maximum power battery temperature	Required
3	1 byte	3087	Maximum temperature detection point number	Required
4	1 byte	3088	Minimum power battery temperature	Required
5	1 byte	3089	Minimum power battery temperature detection point number	Required
6.1	2 people	3090	The voltage of the single power battery is too high/too low (<00>:= normal; <01>:=too high; <10>:=too low)	Required
6.3	2 people	3091	Vehicle power battery state of charge SOC is too high/too low (<00>:= Normal; <01>:=too high; <10>:=too low)	Required
6.5	2 people	3092	Power battery charging overcurrent (<00>:=normal; <01>:=overcurrent flow;<10>:=untrusted state)	Required
6.7	2 people	3093	The power battery temperature is too high (<00>:=normal; <01>:=too high; <10>:=untrusted state)	Required
7.1	2 people	3094	Power battery insulation status (<00>:=normal; <01>:=abnormal Normal; <10>: = untrusted state)	Required
7.3	2 people	3095	Power battery pack output connector connection status (<00>:= normal; <01>:=Abnormal; <10>:=Untrusted state)	Required
7.5	2 people	3096	Charging allowed (<00>:=prohibited; <01>:=allowed)	Required

in:

1) SPN3085 The number of the highest single power battery voltage

Data resolution: 1/bit, 1 offset; data range: 1~256;

2) SPN3086 Maximum power battery temperature

Data resolution: 1 ÿ/bit, -50 ÿ offset; Data range: -50 ÿ ~+200 ÿ;

3) SPN3087 Maximum temperature detection point number

Data resolution: 1/bit, 1 offset; data range: 1~128;

4) SPN3088 minimum power battery temperature

Data resolution: 1 ÿ/bit, -50 ÿ offset; Data range: -50 ÿ ~+200 ÿ;

5) SPN3089 minimum temperature detection point number

Data resolution: 1/bit, 1 offset; data range: 1~128.

Note: When receiving a BSM message with SPN3090---SPN3095 both as 00 (battery status normal) and SPN3096 as 00 (charging prohibited), the charger Pause charging output; when receiving BSM message in which SPN3090---SPN3095 are all 00 (battery status is normal), and SPN3096 is 01 (allowed When the charger resumes charging and the surge current should meet the requirements of 9.7 in GB/T18487.1. If one of the items in SPN3095 (Battery Status) is abnormal, the charger should stop charging.

10.3.5 PGN5376 Single Battery Voltage Message (BMV)

Message function: voltage value of each power battery. Since the maximum length of the data field of PGN5376 exceeds 8 bytes, it is necessary to use For details on the transmission protocol function, see 6.5. The PGN5376 message format is shown in Table 21.

Table 21 PGN5376 message format

Starting byte or bit	length	SPN	SPN Definition	Sending Options
1	2 bytes	3101	#1 single power battery voltage	Optional
3	2 bytes	3102	#2 single power battery voltage	Optional
5	2 bytes	3103	#3 single power battery voltage	Optional
7	2 bytes	3104	#4 single power battery voltage	Optional
9	2 bytes	3105	#5 single power battery voltage	Optional
11	2 bytes	3106	#6 single power battery voltage	Optional
…………				Optional
509	2 bytes	3355	#255 single power battery voltage	Optional
511	2 bytes	3356	#256 Single power battery voltage	Optional

in:

SPN3101~SPN3356 correspond to the voltage of single power batteries #1~#256 respectively:

1-12 bits: Single power battery voltage, data resolution: 0.01V/bit, 0V offset; data range: 0~24V;

13-16 bits: battery group number, data resolution: 1/bit, 0 offset; data range: 0~15.

Note: If the batteries in the vehicle have group numbers, send them according to the actual group numbers; if there are no group numbers, send them as a group of 256 batteries.

10.3.6 PGN5632 Battery Temperature Message (BMT)

Message function: Power battery temperature. When the data length exceeds 8 bytes, it is necessary to use the transmission protocol function for transmission. For details on the format, see 6.5.

The PGN5632 message format is shown in Table 22.

Table 22 PGN5632 message format

Starting byte or bit	length	SPN	SPN Definition	Sending Options
1	1 byte	3361	Power battery temperature 1	Optional
2	1 byte	3362	Power battery temperature 2	Optional
3	1 byte	3363	Power battery temperature 3	Optional
4	1 byte	3364	Power battery temperature 4	Optional
5	1 byte	3365	Power battery temperature 5	Optional
6	1 byte	3366	Power battery temperature 6	Optional
…………				Optional
127	1 byte	3487	Power battery temperature 127	Optional
128	1 byte	3488	Power battery temperature 128	Optional

in:

SPN3361~SPN3488 correspond to the temperature of power battery sampling points 1~128 respectively:

Data resolution: 1°C/bit, -50°C offset; data range: -50°C to +200°C.

10.3.7 PGN5888 Power Battery Reserved Message (BSP)

Message function: Power battery reserved message. When the data field length exceeds 8 bytes, it is necessary to use the transmission protocol function for transmission. For details on the format, see 6.5. The PGN5888 message format is shown in Table 23.

Table 23 PGN5888 message format

Starting byte or bit	length	SPN	SPN Definition	Sending Options
1	1 byte	3491	Power battery reserved field 1	Optional
2	1 byte	3492	Power battery reserved field 2	Optional
3	1 byte	3493	Power battery reserved field 3	Optional
4	1 byte	3494	Power battery reserved field 4	Optional
…………				Optional
16	1 byte	3506	Power battery reserved field 16	Optional

10.3.8 PGN6400BMS Charging Stop Message (BST)

Message function: Let the charger confirm that the BMS will send a charge termination message to terminate the charging process and the reason for termination. The PGN6400 message format is shown in Table 24.

Table 24 PGN6400 message format

Starting byte or bit	length	SPN	SPN Definition	Sending Options
1	1 byte	3511	BMS terminates charging reason	Required
2	2 bytes	3512	BMS charging abort failure reason	Required
4	1 byte	3513	BMS aborts charging error reason	Required

in:

1) Reasons why SPN3511BMS stops charging

1st~2nd place: reach the required SOC target value

<00>:=The required SOC target value was not reached; <01>:=The required SOC target value was reached; <10>:=Not credible state;

3rd to 4th digit: reaches the set value of total voltage

<00>:=The total voltage setting value has not been reached; <01>:=The total voltage setting value has been reached; <10>:=Untrusted state;

5th to 6th digit: reaches the set value of the single cell voltage

<00>:=The single cell voltage setting value has not been reached; <01>:=The single cell voltage setting value has been reached; <10>:=Untrustworthy state;

7th to 8th position: Charger automatically stops

<00>:=Normal; <01>:=Charger aborted (CST frame received); <10>:=Untrusted state.

2) Cause of SPN3512BMS charging failure

1st to 2nd position: insulation fault

<00>:=normal;<01>:=fault;<10>:=untrusted state;

3rd to 4th position: Output connector over-temperature fault

<00>:=normal;<01>:=fault;<10>:=untrusted state;

5th to 6th position: BMS components and output connectors are overheated

<00>:=normal;<01>:=fault;<10>:=untrusted state;

7th to 8th position: Charging connector failure

<00>:=Charging connector normal; <01>:=Charging connector fault; <10>:=Untrusted state;

9th to 10th: Battery pack temperature is too high

<00>:=Battery pack temperature is normal; <01>:=Battery pack temperature is too high; <10>:=Untrusted state;

11th to 12th position: High voltage relay failure

<00>:=normal;<01>:=fault;<10>:=untrusted state;

13th to 14th bit: voltage detection fault at detection point 2

<00>:=normal;<01>:=fault;<10>:=untrusted state;

15th to 16th digit: other faults

<00>:=Normal; <01>:=Fault; <10>:=Untrusted state.

3) SPN3513BMS charging abort error reason

1st to 2nd position: Excessive current

<00>:=current is normal; <01>:=current exceeds the required value; <10>:=untrusted state;

3rd to 4th digit: voltage abnormality

<00>:=Normal; <01>:=Abnormal voltage; <10>:=Untrusted state.

10.3.9 PGN6656 Charger Stop Charging Message (CST)

This message allows the BMS to confirm that the charger is about to end charging and the reason for the end. The PGN6656 message format is shown in Table 25.

Table 25 PGN6656 message format

Starting byte or bit length		SPN	SPN Definition	Sending Options
1	1 byte	3521	Charger stopped charging reason	Required
2	2 bytes	3522	Charger suspends charging fault reason	Required
4	1 byte	3523	Charger aborts charging error reason	Required

in:

1) SPN3521 Charger stops charging reason

1st to 2nd position: The charger stops when the conditions set by the charger are met

<00>:=Normal; <01>:=Charger setting conditions reached and stopped; <10>:=Untrusted state;

3rd to 4th place: Manual termination

<00>:=Normal;<01>:=Manually terminated;<10>:=Untrusted state;

5th to 6th position: Fault abort

<00>:=Normal;<01>:=Fault abort;<10>:=Untrusted state;

7th to 8th position: BMS automatically stops

<00>:=Normal; <01>:=BMS abort (BST frame received); <10>:=Untrusted state.

2) SPN3522 Charger stops charging fault reason

1st~2nd position: Charger overtemperature fault

<00>:=charger temperature is normal; <01>:=charger overtemperature; <10>:=untrusted state;

3rd to 4th position: Charging connector failure

<00>:=Charging connector normal; <01>:=Charging connector fault; <10>:=Untrusted state;

5th to 6th position: Charger internal over-temperature fault

<00>:=charger internal temperature is normal; <01>:=charger internal temperature is over; <10>:=untrusted state;

7th to 8th digit: The required amount of power cannot be transmitted

<00:=Power transmission is normal; <01>:Power cannot be transmitted; <10>:=Untrusted state;

9th to 10th digit: Charger emergency stop failure

<00>:=Normal;<01>:=Charger emergency stop;<10>:=Untrusted state;

11th to 12th: Other faults

<00>:=Normal; <01>:=Fault; <10>:=Untrusted state.

3) SPN3523 Charger aborts charging error reason

1st to 2nd digit: Current mismatch

<00>:=current matching; <01>:=current mismatch; <10>:=untrusted state;

3rd to 4th digit: voltage abnormality

<00>:=Normal; <01>:=Abnormal voltage; <10>:=Untrusted state.

10.4 Charging End Phase Message

10.4.1 PGN7168 BMS Statistical Data Message (BSD)

This message allows the charger to confirm the BMS's charging statistics for the current charging process. The PGN7168 message format is shown in Table 26.

Table 26 PGN7168 message format

Starting byte or bit length		SPN	SPN Definition	Sending Options
1	1 byte	3601	Termination state of charge SOC (%)	Required
2	2 bytes	3602	Minimum voltage of power battery cell (V)	Required
4	2 bytes	3603	Maximum voltage of power battery cell (V)	Required
6	1 byte	3604	Power battery minimum temperature (ÿ)	Required
7	1 byte	3605	Maximum temperature of power battery (ÿ)	Required

in:

1) SPN3601 Stop State of Charge SOC

Data resolution: 1%/bit, 0% offset; Data range: 0~100%;

2) SPN3602 Minimum voltage of power battery cells

Data resolution: 0.01V/bit, 0V offset; Data range: 0 ~ 24V;

3) SPN3603 power battery cell maximum voltage

Data resolution: 0.01V/bit, 0V offset; Data range: 0 ~ 24V;

4) SPN3604 Minimum power battery temperature

Data resolution: 1 ÿ/bit, -50 ÿ offset; Data range: -50 ÿ ~+200 ÿ;

5) SPN3605 Maximum power battery temperature

Data resolution: 1°C/bit, -50°C offset; data range: -50°C to +200°C.

10.4.2 PGN7424 Charger Statistics Data Message (CSD)

Message function: Confirm the charging statistics of the charger during the current charging process. The PGN7424 message format is shown in Table 27.

Table 27 PGN7424 message format

Starting byte or bit length		SPN	SPN Definition	Sending Options
1	2 bytes	3611	Accumulated charging time (min)	Required
3	2 bytes	3612	Output energy (kW·h)	Required
5	4 bytes	3613	Charger number, 1/bit, 1 offset, data range: 0~0xFFFFFFFF	Required

in:

1) SPN3611 cumulative charging time

Data resolution: 1min/bit, 0min offset; data range: 0~600min;

2) SPN3612 output energy

Data resolution: 0.1kW·h/bit, 0kW·h offset; data range: 0~1000kW·h.

10.5 Error Messages

10.5.1 PGN7680 BMS Error Message (BEM)

Message function: When BMS detects an error, it sends a charging error message to the charger until BMS receives the charging error message sent by the charger.
The machine identification message (CRM) is displayed or the charging plug is unplugged. The PGN7680 message format is shown in Table 28.

Table 28 PGN7680 message format

Starting byte or bit length		SPN	SPN Definition	Sending Options
1.1	2 people	3901	Receiving the charger identification message with SPN2560=0x00 timed out (<00>: = normal; <01>: = timeout; <10>: = untrusted state)	Required
1.3	2 people	3902	Receiving the charger identification message with SPN2560=0xAA timed out (<00>: = normal; <01>: = timeout; <10>: = untrusted state)	Required
2.1	2 people	3903	Receive the charger time synchronization and charger maximum output capacity message When (<00>:=normal;<01>:=timeout;<10>:=untrusted state)	Required
2.3	2 people	3904	Receiving the charger's charging preparation message timeout (<00>:= normal; <01>:=timeout;<10>:=untrusted state)	Required
3.1	2 people	3905	Receiving the charger charging status message timeout (<00>:= normal; <01>:=timeout;<10>:=untrusted state)	Required
3.3	2 people	3906	Receiving charger stop charging message timeout (<00>:= normal; <01>:=timeout;<10>:=untrusted state)	Required
4.1	2 people	3907	Receiving the charging statistics message from the charger timed out (<00>:= normal; <01>:=timeout;<10>:=untrusted state)	Required
4.3	6-digit		other	Optional

10.5.2 PGN7936 Charger Error Message (CEM)

Message function: When the charger detects an error, it sends a charging error message to the BMS until the charger receives the message sent by the BMS.

The BRM message is sent or the charging plug is unplugged. The PGN7936 message format is shown in Table 29.

Table 29 PGN7936 message format

Starting byte or bit length		SPN	SPN Definition	Sending Options
1.1	2 people	3921	Receiving the identification message from BMS and vehicle timed out (<00>:= normal; <01>:=timeout;<10>:=untrusted state)	Required
2.1	2 people	3922	Receiving battery charging parameter message timeout (<00>:=normal; <01>: =Timeout; <10>:=Untrusted state)	Required
2.3	2 people	3923	Receiving BMS charging preparation message timeout (<00>:= normal; <01>:=timeout;<10>:=untrusted state)	Required
3.1	2 people	3924	Timeout in receiving battery charging status message (<00>:= normal; <01>:=timeout;<10>:=untrusted state)	Required
3.3	2 people	3925	Receiving battery charging request message timeout (<00>:=normal; <01>: =Timeout; <10>:=Untrusted state)	Required
3.5	2 people	3926	Receiving BMS charging abort message timeout (<00>:=normal; <01>: =Timeout; <10>:=Untrusted state)	Required
4.1	2 people	3927	Receiving BMS charging statistics message timeout (<00>:=normal; <01>: =Timeout; <10>:=Untrusted state)	Required
4.3	6-digit		other	Optional

Appendix A

(Informative Appendix)

Charging Process

A.1 Charging working state transition

After the BMS and charger are physically connected and powered on, the state transition between the BMS and charger is an interoperability convention for coordinated operation. Typical charging state transitions are shown in Figures A.1 to A.5.

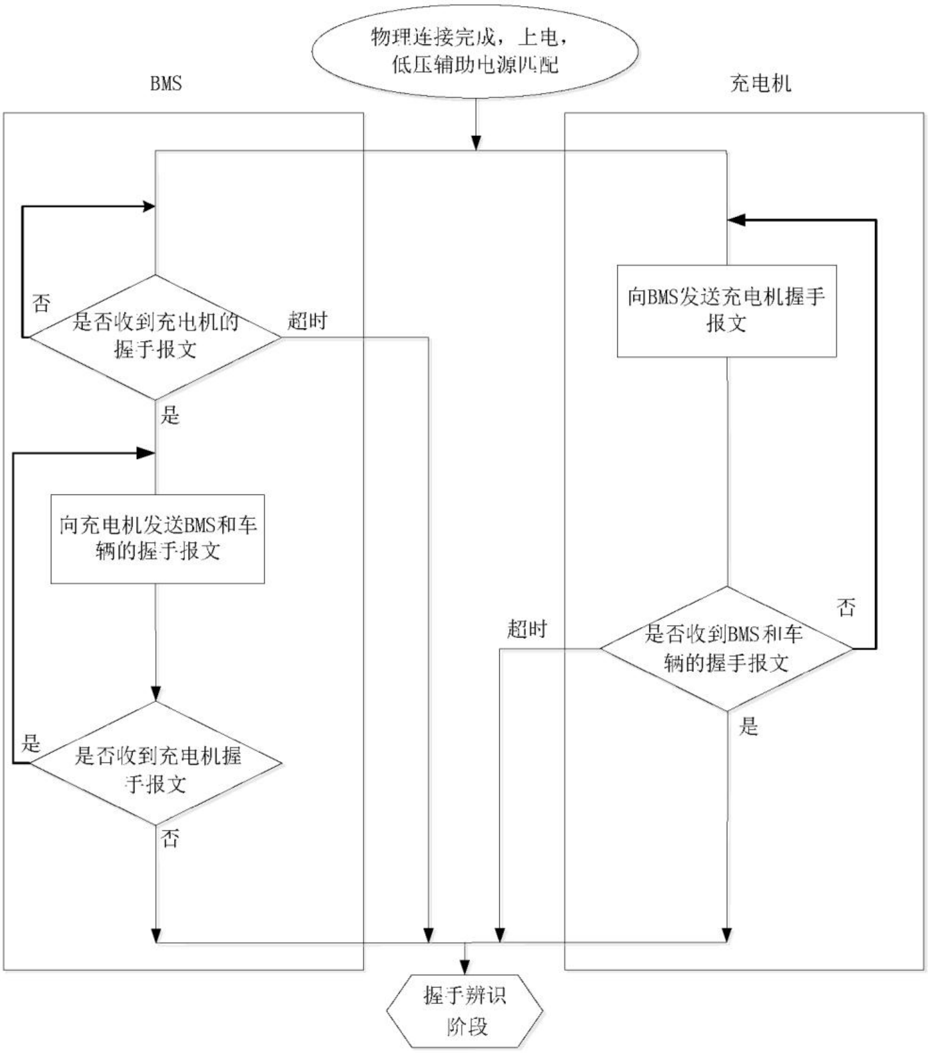

Figure A.1 Charging handshake startup flow chart

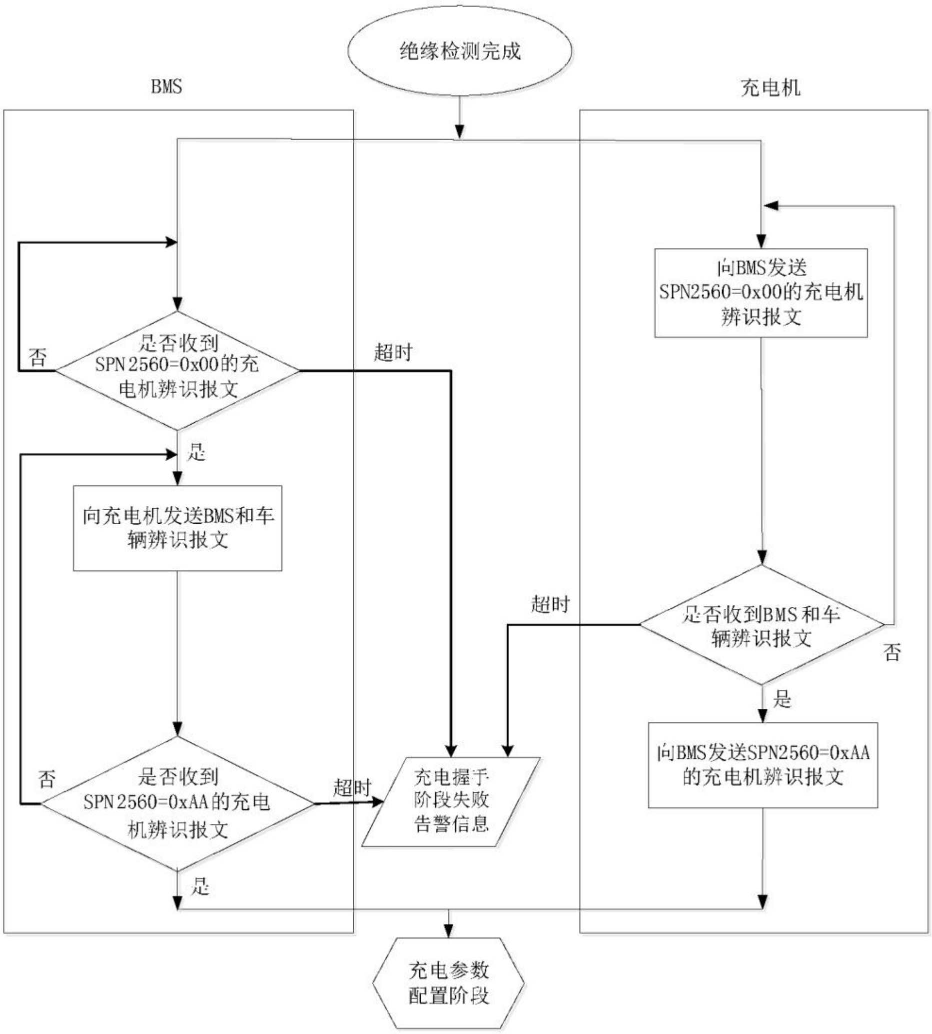

Figure A.2 Charging handshake identification flow chart

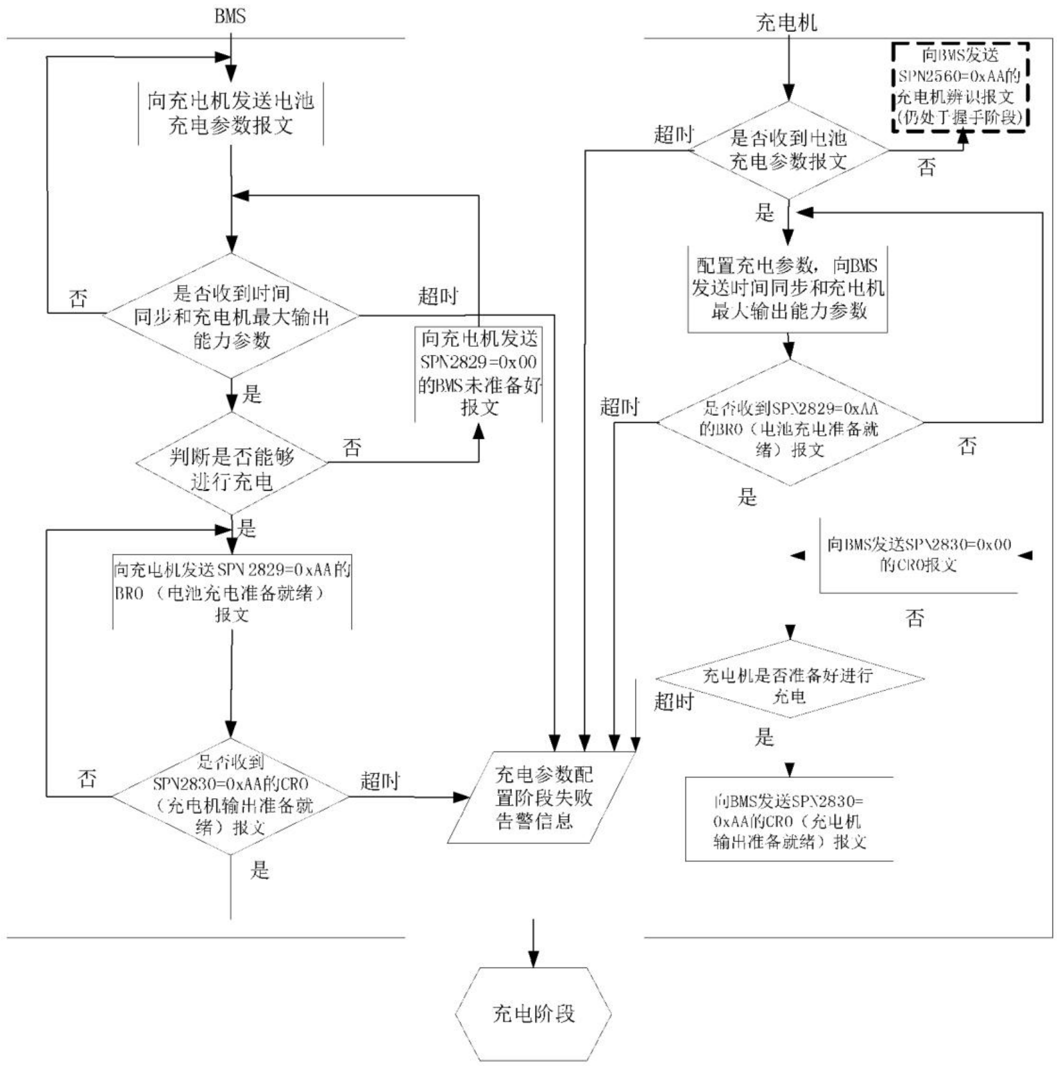

Figure A.3 Flowchart of the charging parameter configuration phase

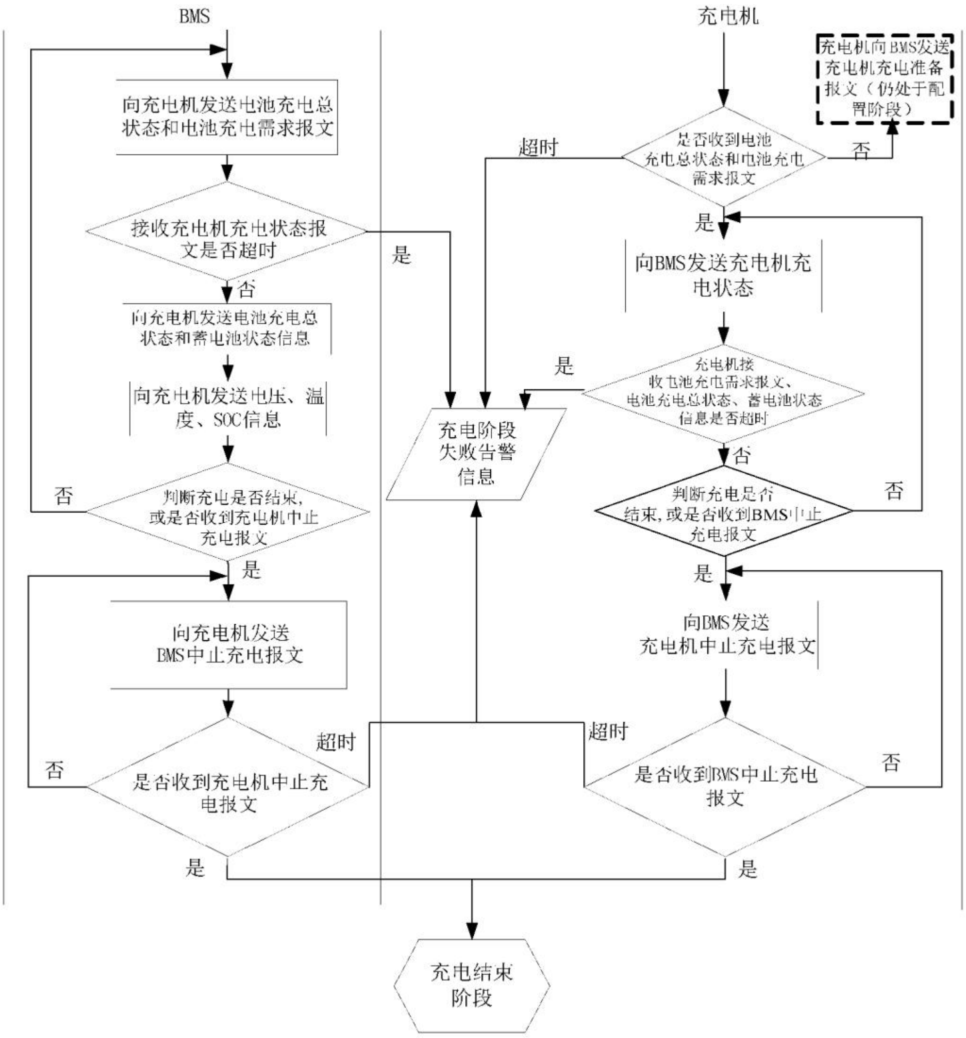

Figure A.4 Charging stage flow chart

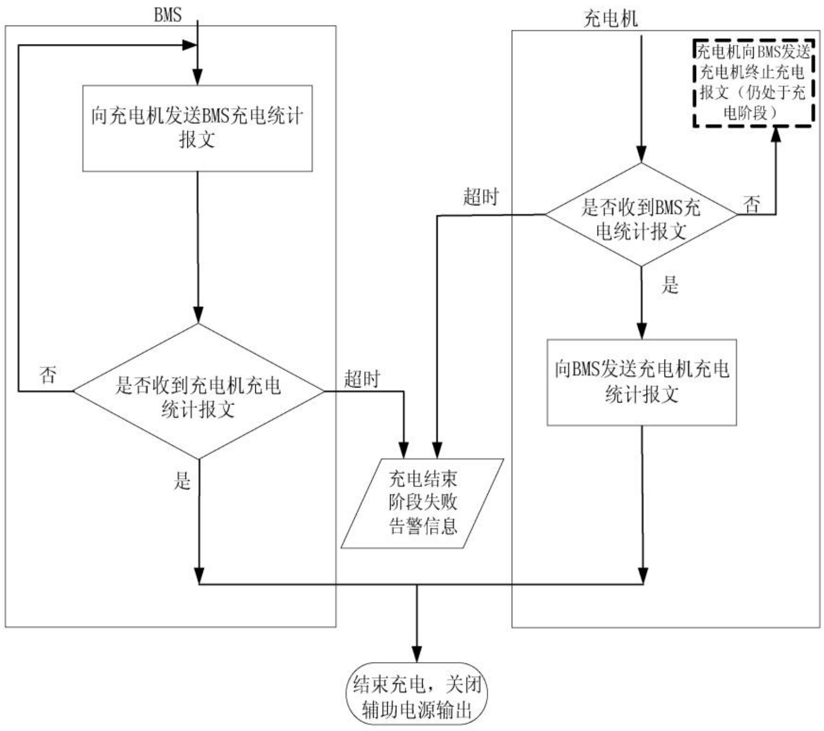

Figure A.5 Flowchart of the charging end stage

A.2 Charging Sequence Flowchart

The detailed charging sequence flow chart is shown in Figures A.6 to A.12.

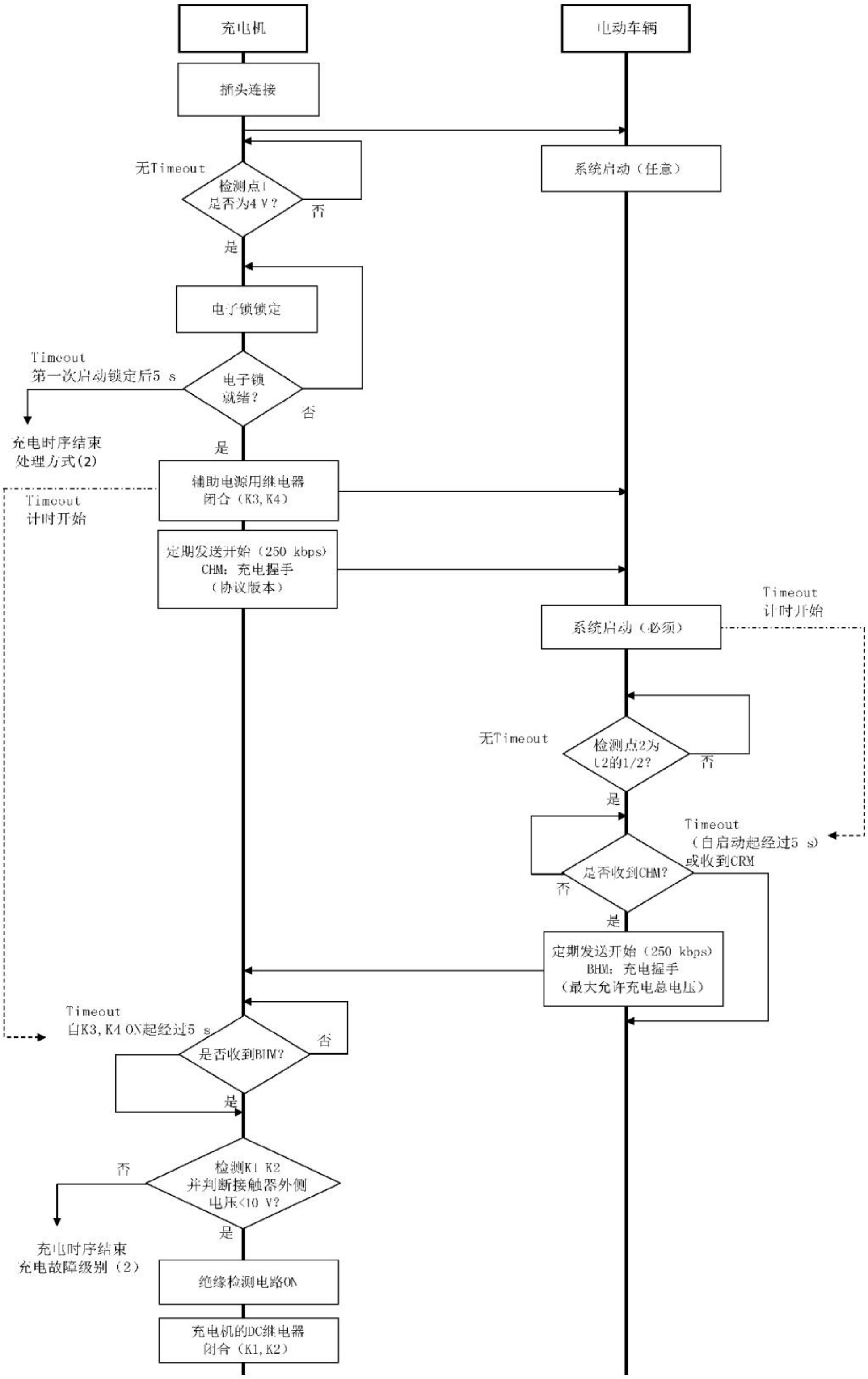

Figure A.6 Normal charging sequence flow chart

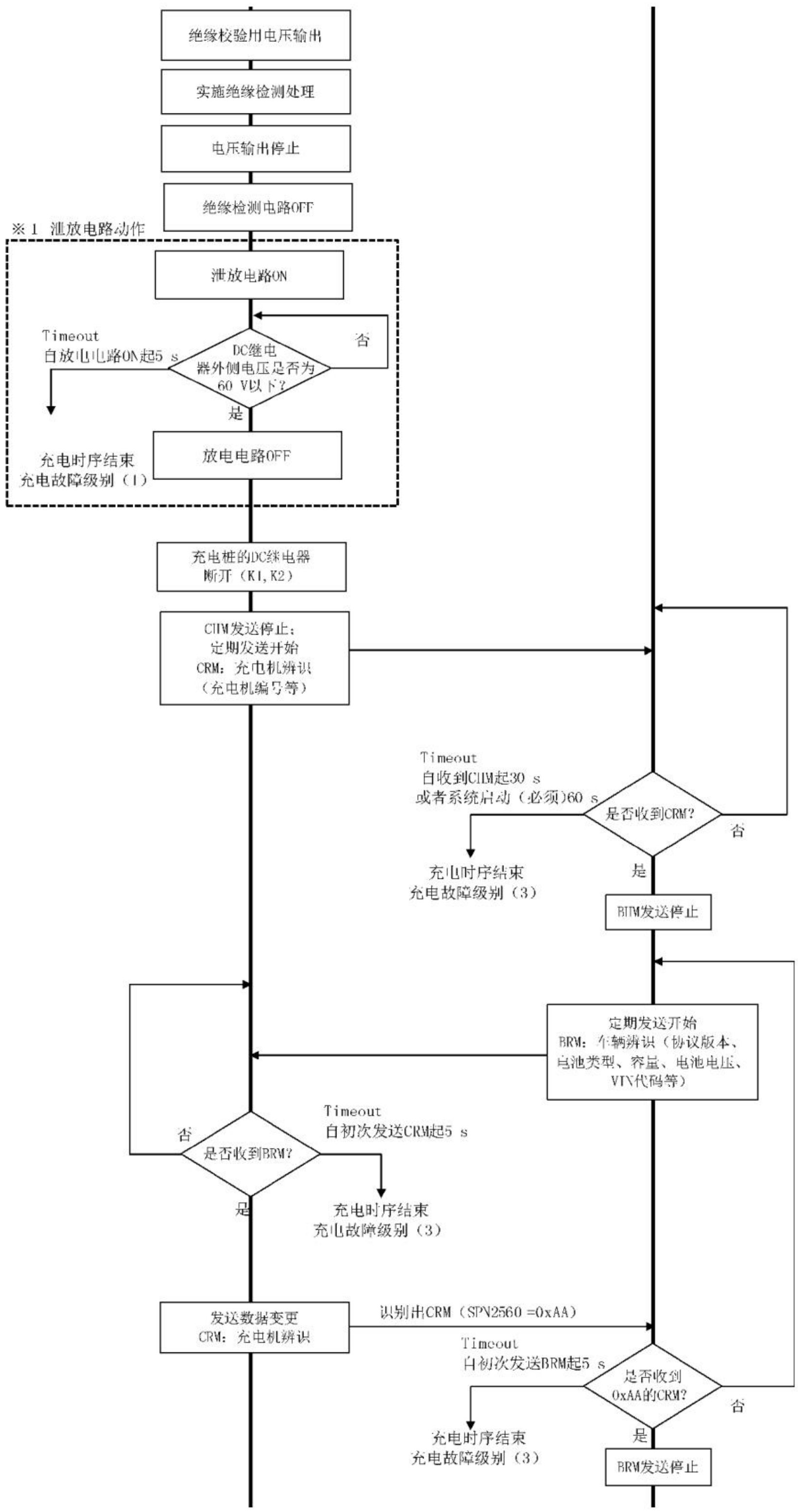

Figure A.6 (continued)

Machine Translated by Google

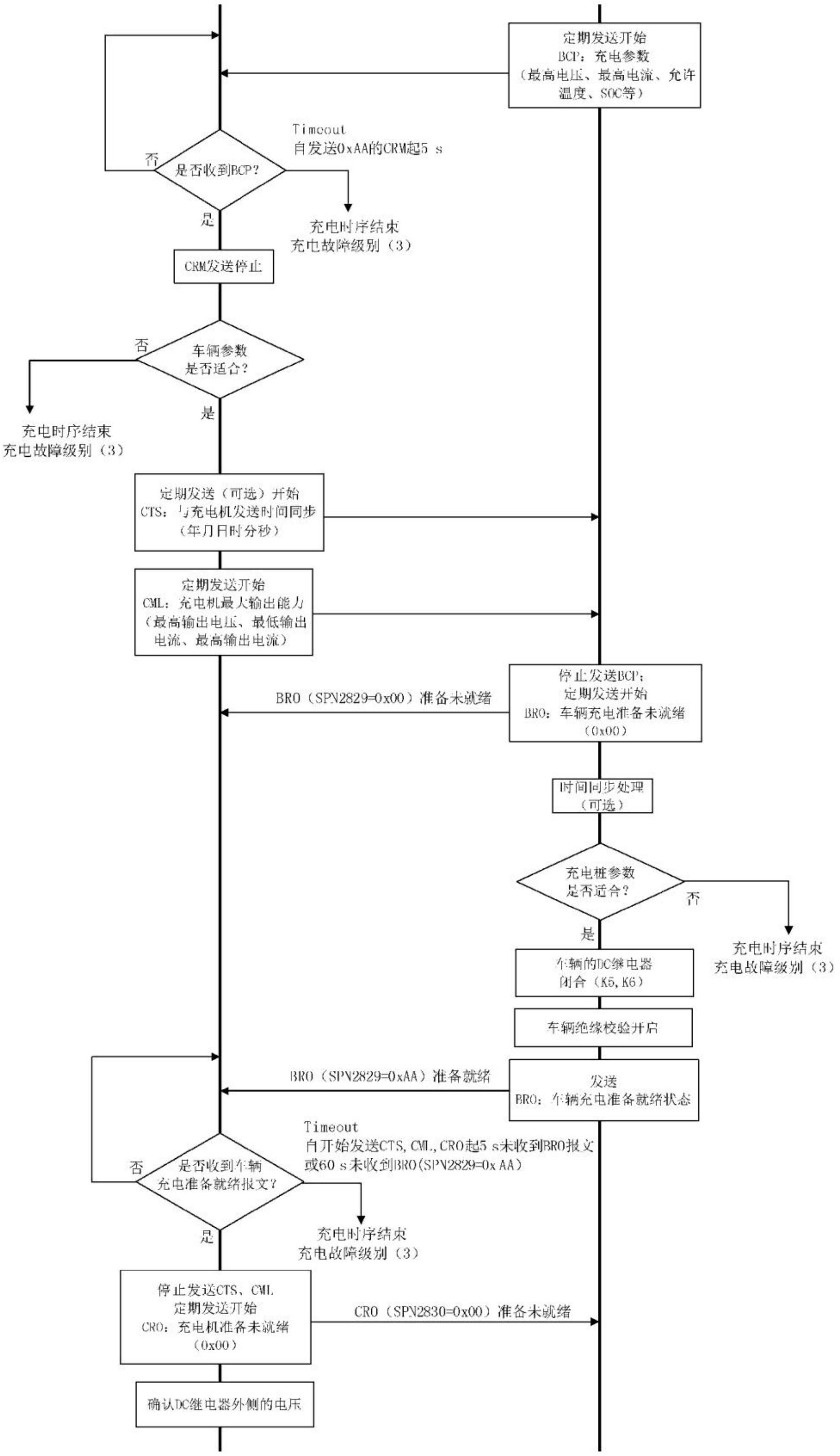

Figure A.6 (continued)

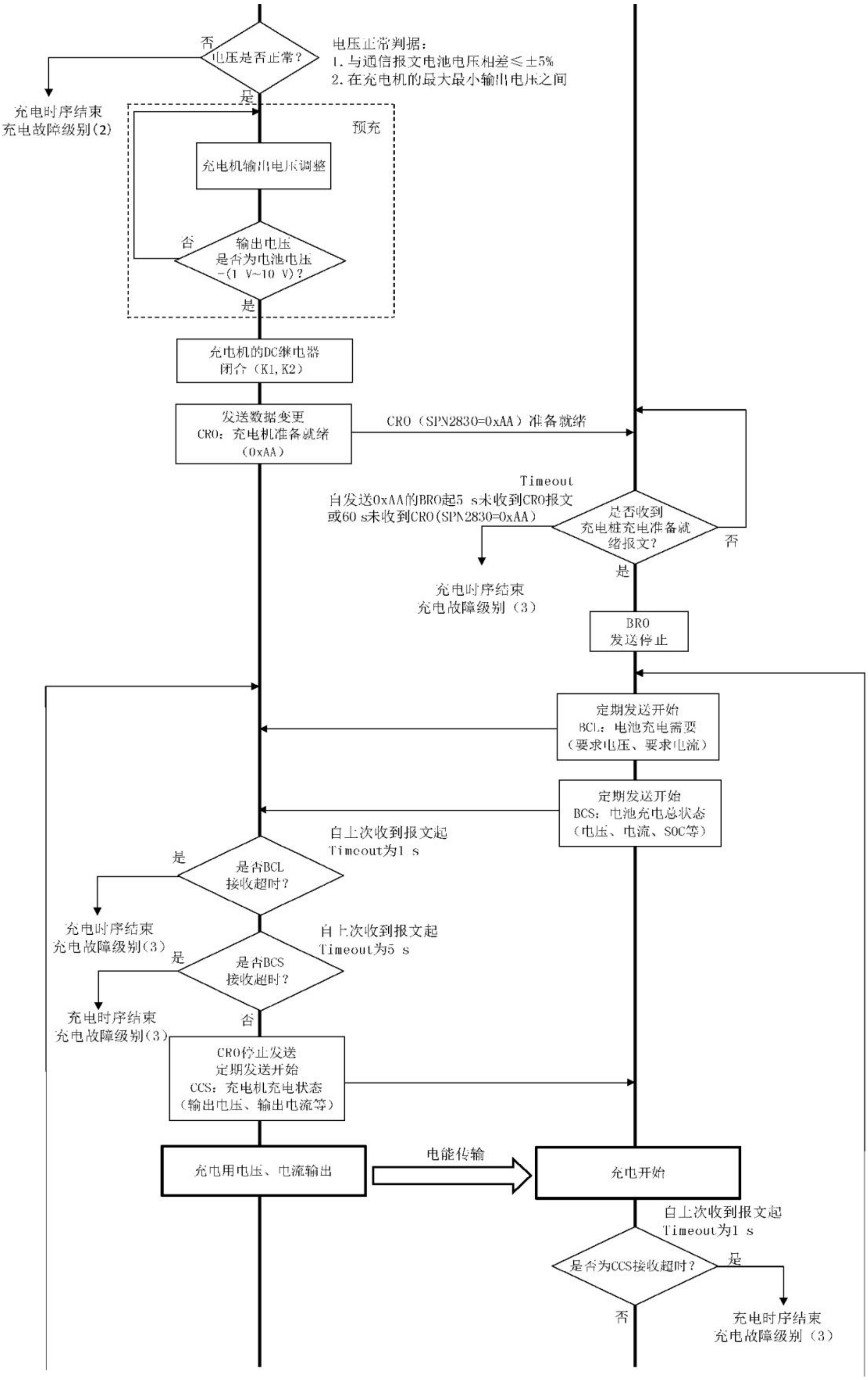

Figure A.6 (continued)

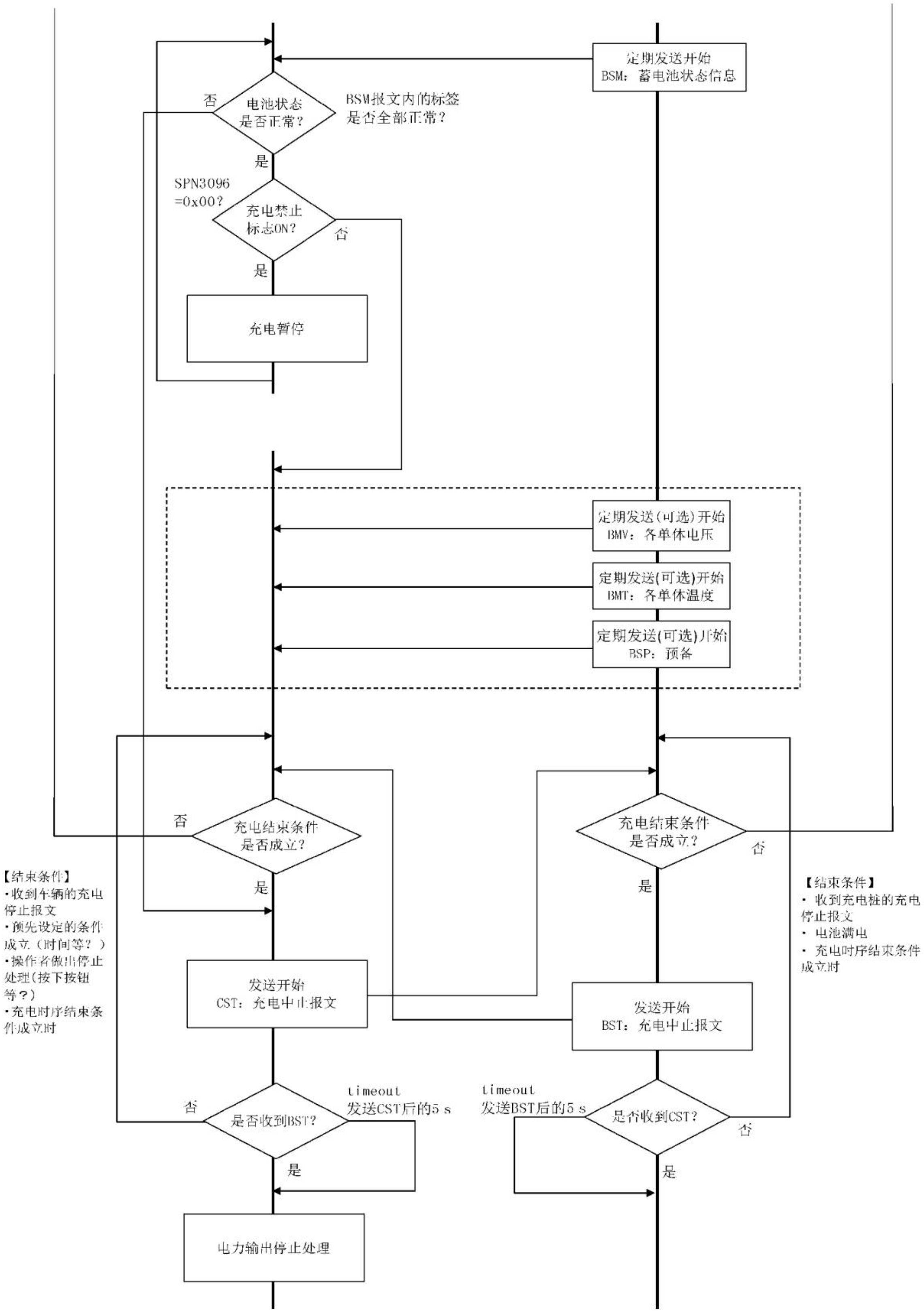

Figure A.6 (continued)

Machine Translated by Google

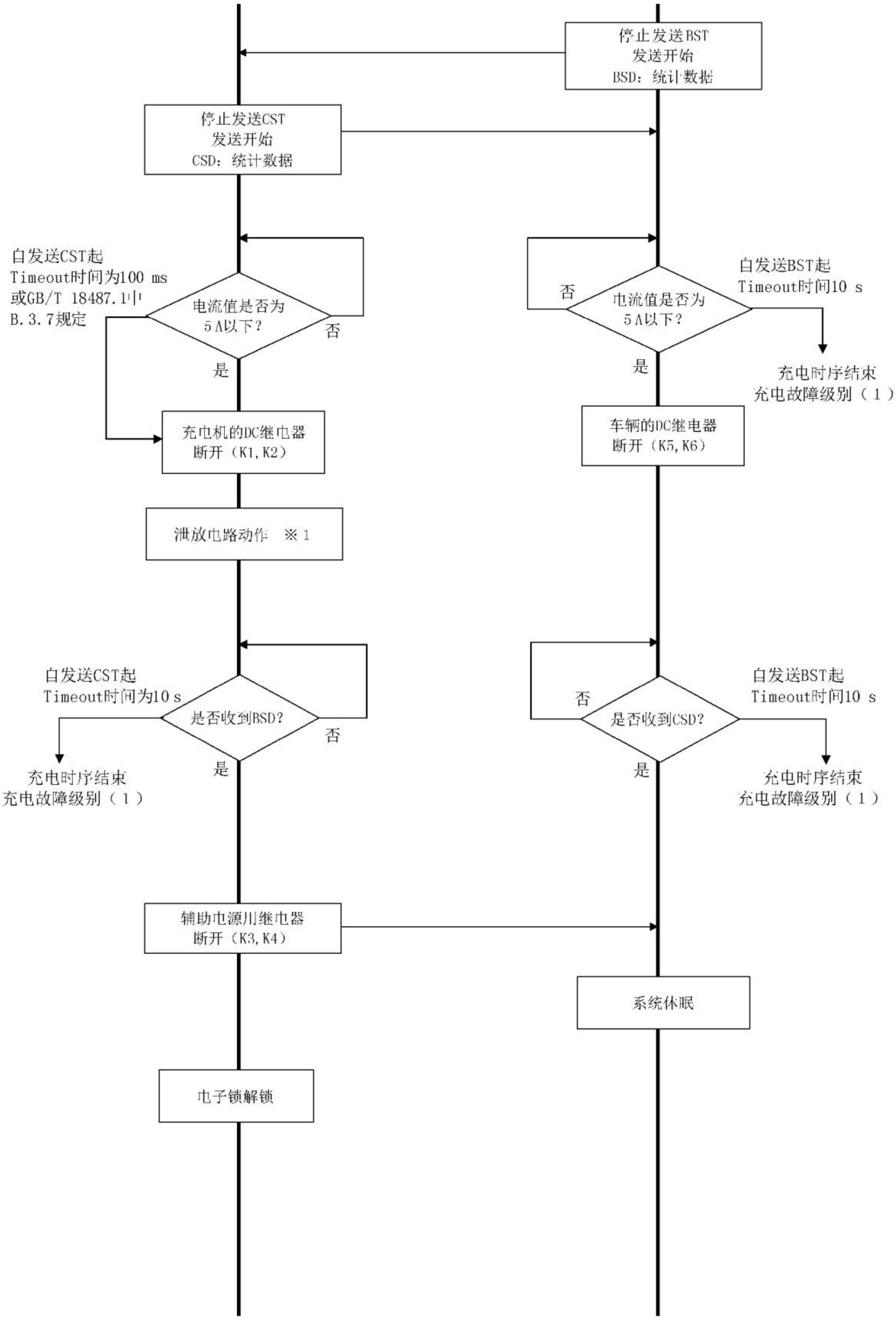

Figure A.6 (continued)

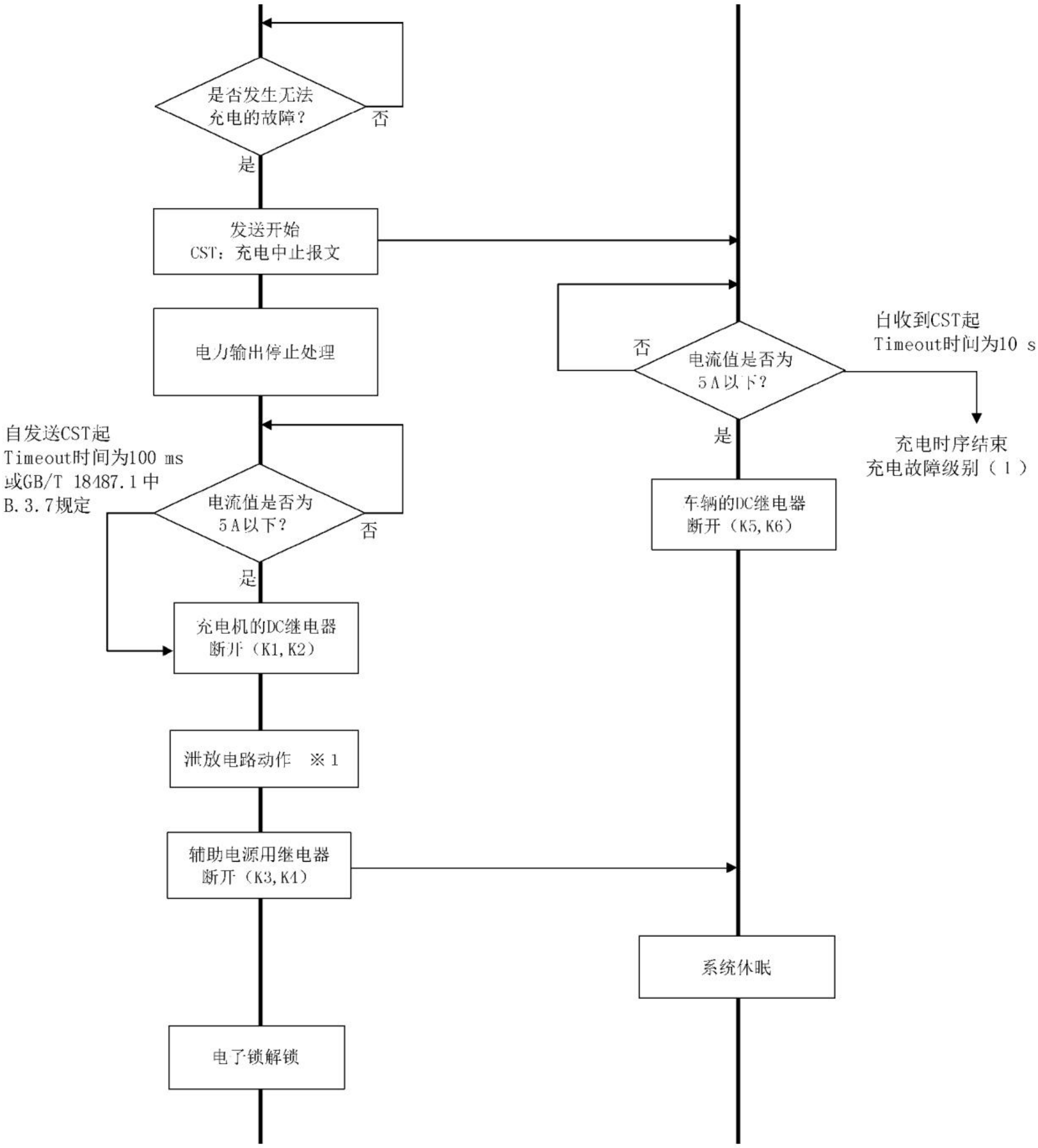

Figure A.7 Flowchart of stopping in abnormal state (charger-related reasons)

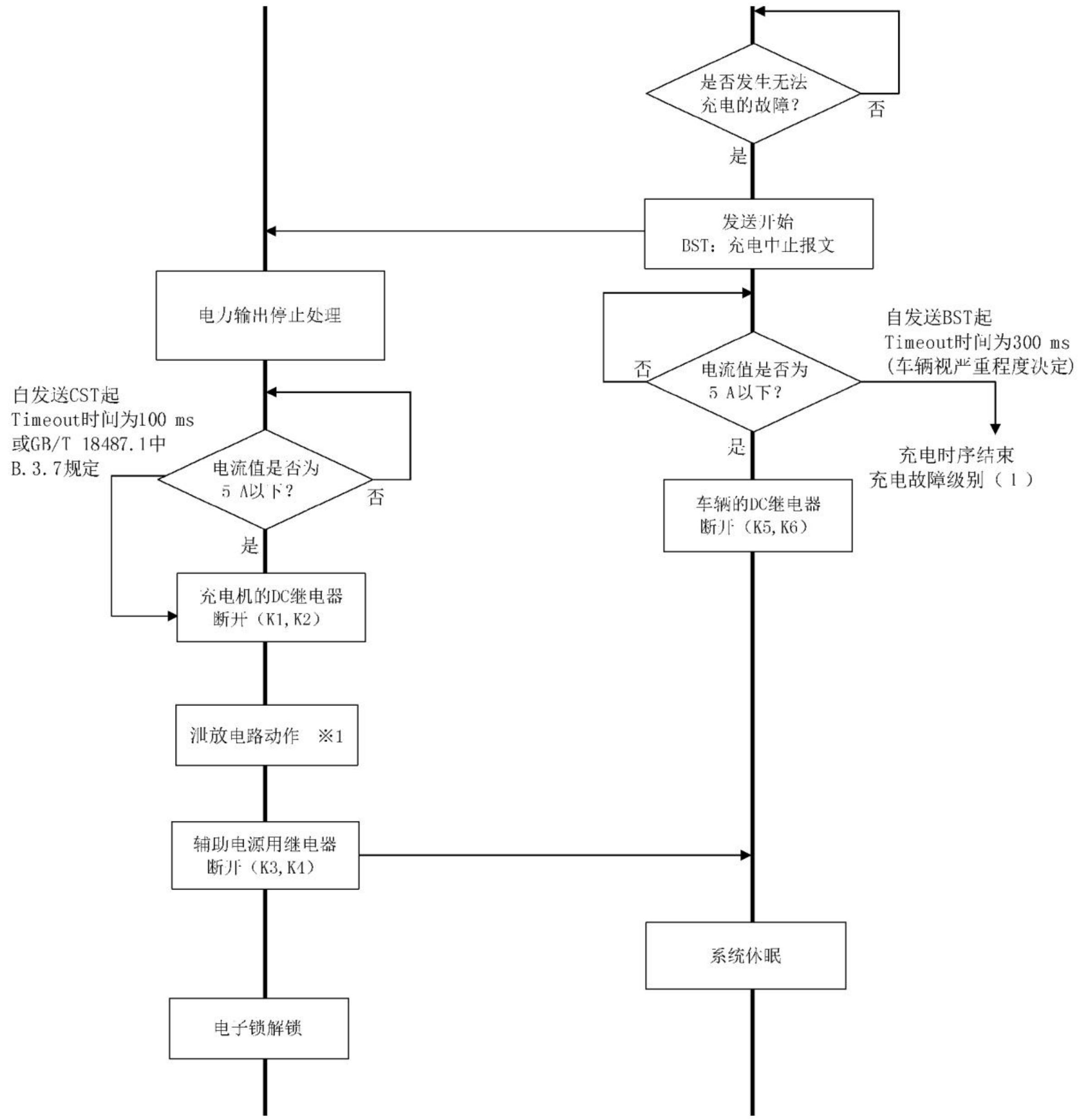

Figure A.8 Flowchart of stopping in abnormal state (cause on vehicle side)

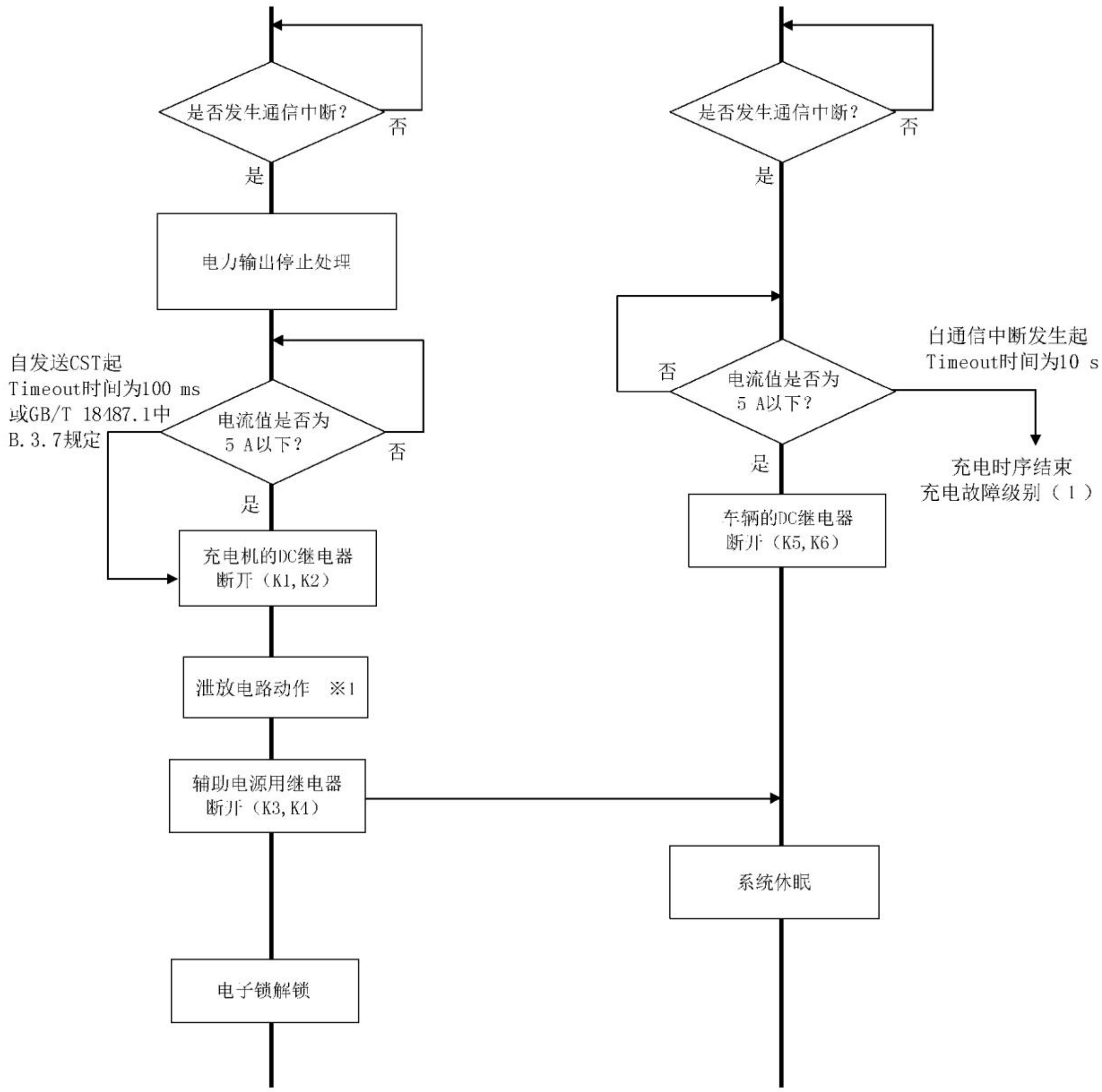

Figure A.9 Communication interruption: Communication timeout occurs after 3 attempts of reconnection and communication abort occurs.

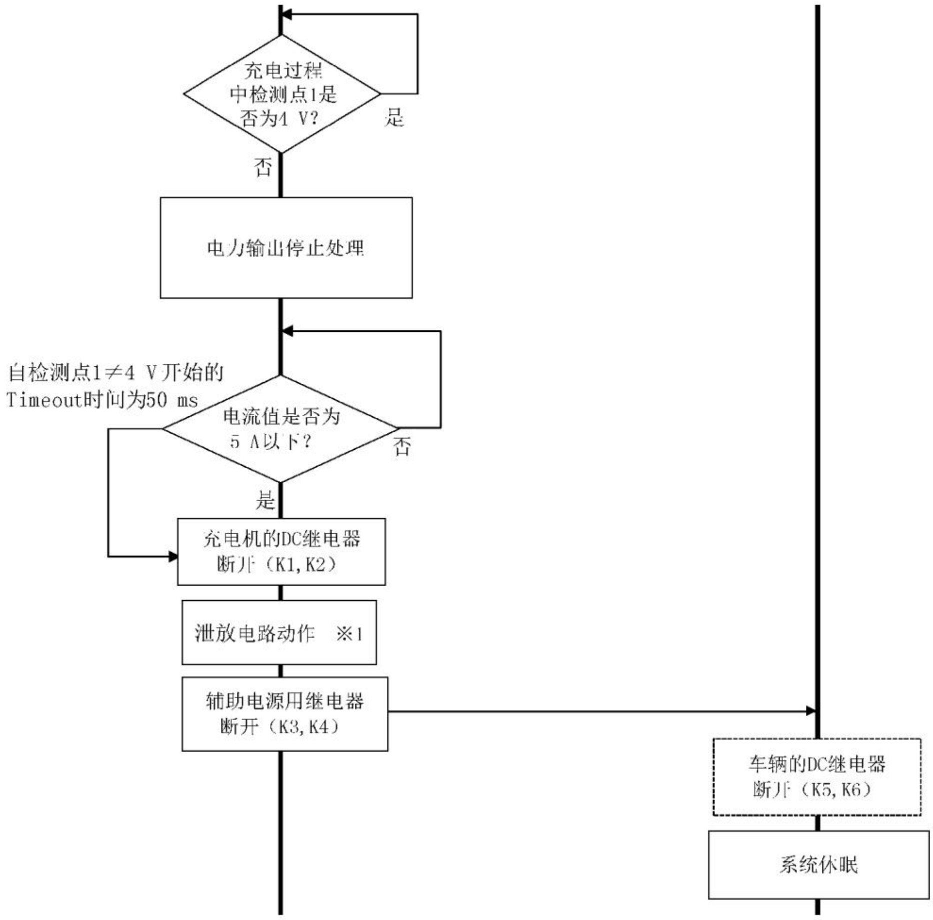

Figure A.10 S switch opening flow chart

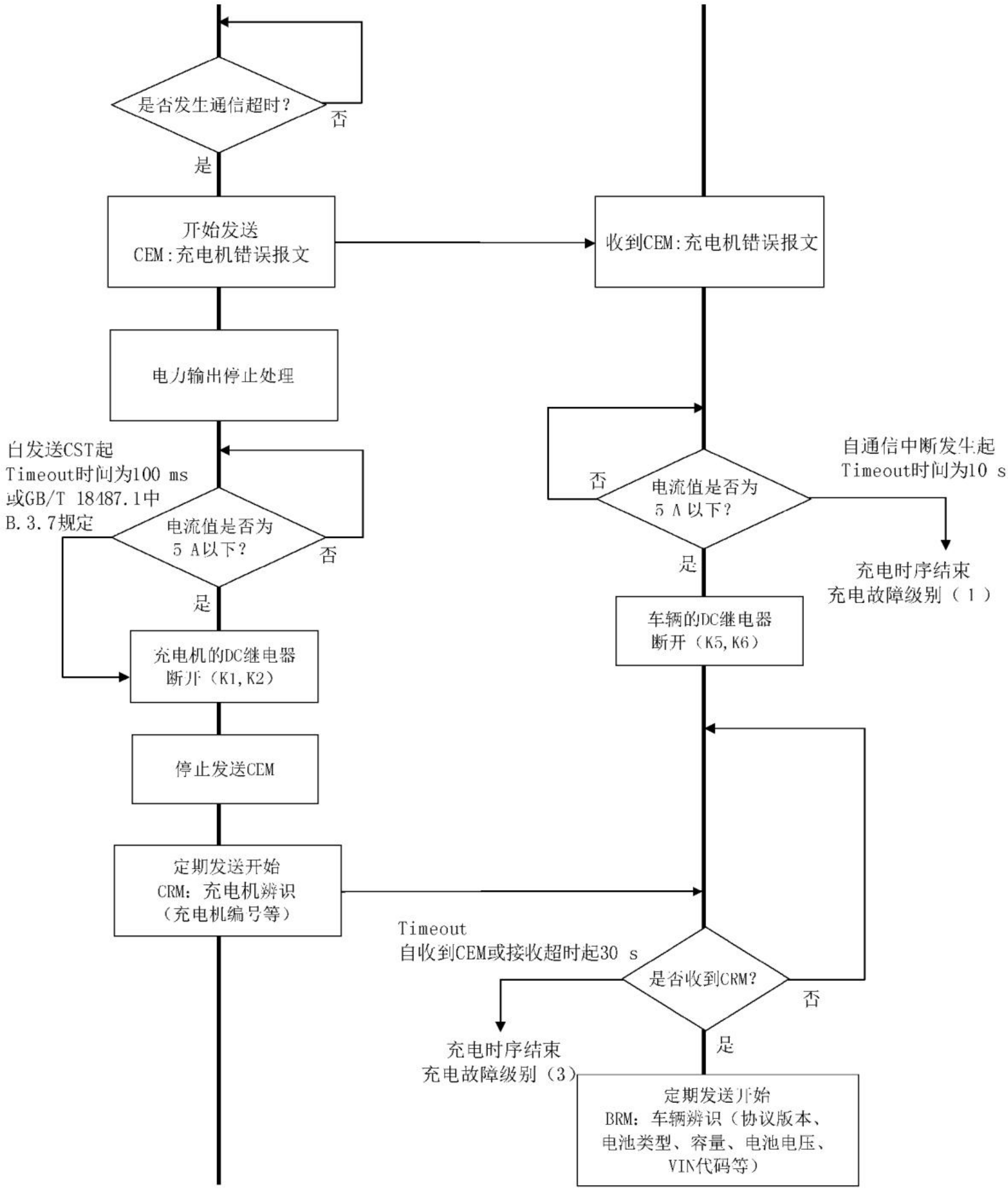

Figure A.11 Flowchart of Charger Receiving BMS Communication Timeout

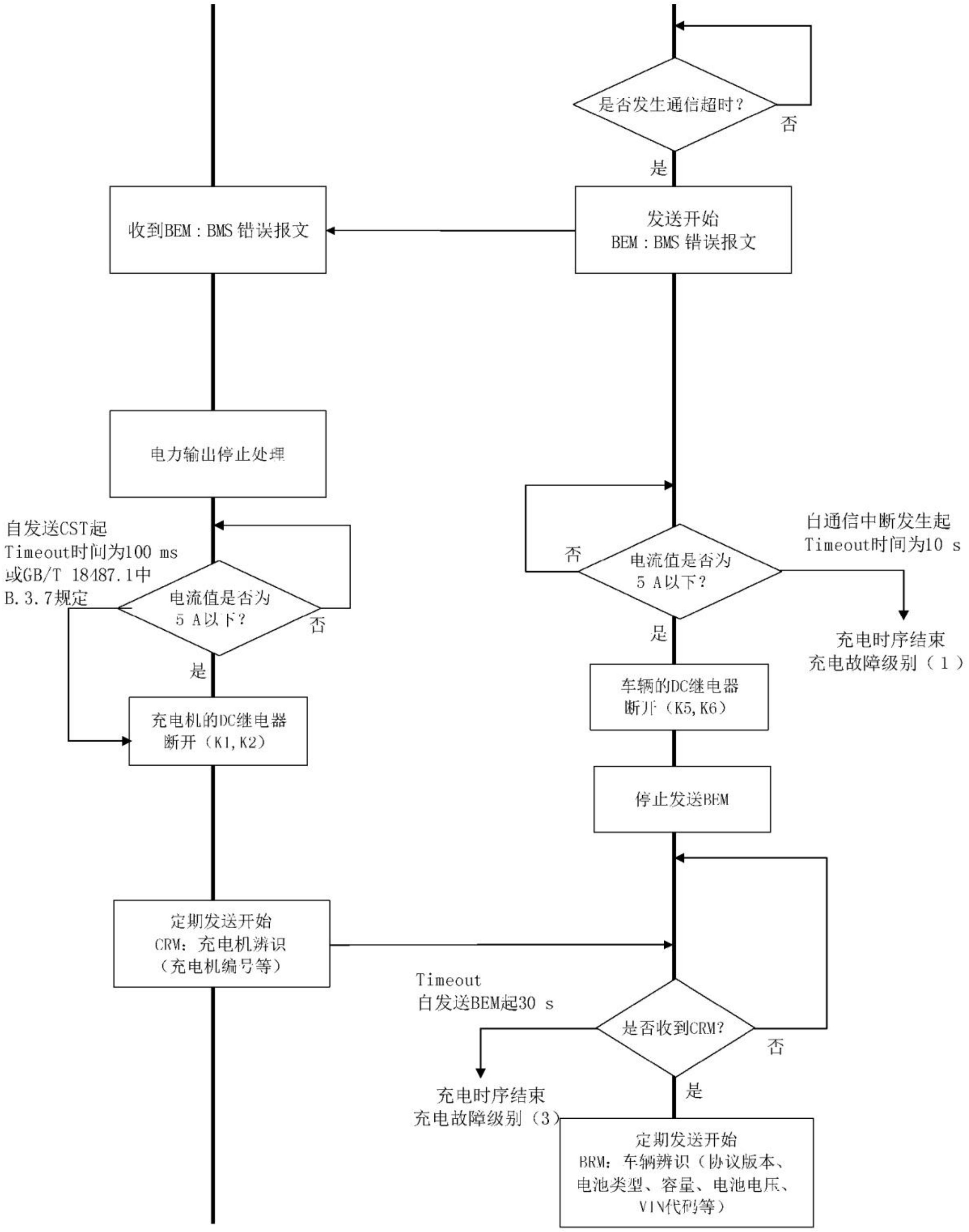

Figure A.12 Flowchart of BMS receiving charger communication timeout

Appendix B

(Informative) Charger

and BMS Fault Diagnosis Messages

B.1 Diagnostic Trouble Codes

The Diagnostic Trouble Code (DTC) consists of four independent fields, which are shown in Table B.1:

Table B.1 Diagnostic Trouble Codes (DTCs)

Serial number	Independent domain
1	Suspect Parameter Number (SPN) where the fault occurred (19 bits)
2	Fault Mode Identifier (FMI) (5 bits)
3	Occurrence count (OC) (7 digits)
4	Conversion method (CM) of suspicious parameter number (1 digit)

The 19-digit Suspect Parameter Number (SPN) identifies the diagnostic item in the fault report. The Suspect Parameter Number is unrelated to the address code of the control module that sends the fault diagnostic information. The SPN number is defined in Section 10.3 for BMS and charger hardware fault information, such as SPN3090-SPN3095, SPN3511-SPN3513, and SPN3521-SPN3523.

The Fault Mode Identifier (FMI) defines the type of fault found in the BMS and charger. Its data length is 5 bits and the data status is 0~

There are 31 total 32 types. The currently defined failure mode identifiers are as follows:

<0>: = Power battery voltage fault; <1>: = Power battery

current fault; <2>: = Power battery temperature fault;

<3>: = Power battery insulation status; <4>: = Power

battery output connector overtemperature fault; <5>: =

BMS component, battery pack output connector overtemperature; <6>: =

Charger temperature fault; <7>: = Charger connector fault; <8>: = Charger internal

temperature fault; <9-31>: = Reserved for

backup. The Occurrence Count (OC) defines the

number of times a fault changes from its previous active

state to its current active state. The maximum

value is 126. The count overflows.

When an event occurs, the counter value remains at 126. If the number of occurrences is unknown, all bits in this field are set to 1.

The Conversion Mode (CM) of the questionable parameter number is set to 0, indicating that the SPN bits are all in Intel format.

B.2 Fault Diagnosis Message Classification

The classification of fault diagnosis messages is shown in Table B.2.

Table B.2 Fault diagnosis message classification

Message code name	Message Description	PGN	PGN (Hex)	priority	data length	Message cycle
DM1	Current fault code	8192	002000H	6	indefinite	Incident Response
DM2	Historical fault codes	8448	002100H	6	indefinite	Incident Response
DM3	Diagnostic Readiness	8704	002200H	6	2 bytes	Incident Response
DM4	Clear/reset current fault code	8960	002300H	6	0	Incident Response
DM5	Clearing/resetting historical fault codes	9216	002400H	6	0	Incident Response
DM6	Freeze Frame Parameters	9472	002500H	6	indefinite	Incident Response

B.3 Fault Diagnosis Message Format and Content

Fault diagnosis message and content include:

a) PGN8192 diagnostic information 1, current fault code message (DM1)

Message function: When a fault occurs, send the current fault code. Each fault code is 4 bytes. The data segment exceeds 8 bytes using the transmission Protocol function transmission, the format is detailed in 6.5. The PGN8192 message format is shown in Table B.3.

Table B.3 PGN8192 message format

Start byte or bit	length	definition
1	1-byte	lower 8 significant bits of the first current fault code SPN
2	1 byte	The second byte of the first current fault code SPN
3.1	The upper 3 digits	of the first 3-digit current fault code SPN
3.4	5-bit	fault mode flag, see B.1 for details
4.1	7-digit	occurrence count
4.8	1-bit	suspicious parameter number conversion method, set to 0
…………		

b) PGN8448 Diagnostic Information 2, Historical DTC Message (DM2)

Message function: This data includes a series of diagnostic codes and the number of occurrences of historical fault codes. Each fault code is 4 bytes. The excess 8 bytes in the data segment are transmitted using the transport protocol function. The format is detailed in 6.5. The PGN8448 message format is shown in Table B.4.

Table B.4 PGN8448 message format

Start byte or bit	length	definition
1	1-byte	lower 8 significant bits of the first historical fault code SPN
2	1 byte	The second byte of the first historical fault code SPN
3.1	The high 3 digits	of the first 3-digit historical fault code SPN
3.4	5-bit	fault mode flag, see B.1 for details

Table B.4 (continued)

Start byte or bit	length	definition
4.1	7-digit occurrence count	
4.8	1-bit suspicious parameter number conversion method, set to 0	
…………		

c) PGN8704 Diagnostic Information 3, Diagnostic Readiness Message (DM3)

Message function: Reports diagnostic information related to diagnostic readiness. The PGN 8704 message format is shown in Table B.5.

Table B.5 PGN8704 message format

Start byte or bit	length	definition
1	1 byte current fault code number	
2	1-byte number of historical fault codes	

d) PGN8960 Diagnostic Information 4, Clear/Reset Message for Current DTC (DM4)

Message function: All diagnostic information about the current fault code should be cleared. When you need to clear the diagnostic information related to the current fault code, When the operation is completed or there is no fault code in the requested control module, the control module is requested to Send a positive response. If for some reason the control module cannot perform the requested operation, it must send a negative response. The information related to the current fault code includes: the number of current fault codes, diagnostic readiness status information and the current fault code.

e) PGN9216 Diagnostic Information 5, Clear/Reset Historical Fault Code Message (DM5)

Message function: When a control module receives a request command for this parameter group, all diagnostic information related to historical fault codes should be If the diagnostic data related to the current fault code is cleared, it will not be affected. If there is no historical fault code, a positive response must be sent. The reason is that the control module cannot execute the request instruction of this parameter group, so it must send a negative response. Related information includes: the number of historical fault codes, diagnostic readiness status information and historical fault codes.

f) PGN9472 Diagnostic Information 6, Freeze Frame Parameter Message (DM6)

Message function: When a diagnostic fault code is received, a series of parameters are recorded. Each fault code is 4 bytes. The data segment is more than 8 The bytes are transmitted using the transport protocol function. The format is specified in 6.5. The PGN9472 message format is shown in Table B.6.

Table B.6 PGN9472 message format

Start byte or bit	length	definition
1	1-byte freeze frame length of the first fault diagnostic code	
2	1-byte lower 8 significant bits of the first fault diagnostic code SPN	
3	1 byte The second byte of the first diagnostic fault code SPN	
4.1	The upper 3 digits of the first 3-digit diagnostic trouble code SPN	
4.4	5-bit fault mode flag, see B.1 for details	
5.1	7-digit occurrence count	
5.8	1-bit suspicious parameter number conversion method, set to 0	
…………		

Appendix C

(Informative Appendix)

Troubleshooting Methods during Charging

C.1 Troubleshooting

Troubleshooting methods include:

Method a) - Immediately shut down the charger and wait for professional maintenance personnel to

repair it; Method b) - Stop charging and make a fault record (the charging cable must be reconnected before the next charge can be carried out); Method c) -

Suspend charging and automatically resume charging after the fault is eliminated (after detecting that the fault state is resolved, re-communication handshake and start charging).

C.2 Charging Failure Classification and Treatment

The classification and treatment of charging faults are shown in Table C.1.

Table C.1 Charging fault classification and treatment methods

Fault level	Fault classification and handling methods
1	Personal safety level fault classification and handling methods: 1) Insulation fault: Handling method a). 2) Leakage fault: Handling method a). 3) Emergency stop fault: Handling method a)
2	Equipment safety level fault classification and handling methods: 1) Connector failure (fault detected by pilot circuit): Processing method b). 2) BMS components and output connectors are overheated: Processing method b). 3) Battery pack temperature is too high: Solution b). 4) Battery cell voltage is too low or too high: Processing method b). 5) BMS detects that the charging current is too large or the charging voltage is abnormal: Processing method b). 6) The charger detects that the charging current does not match or the charging voltage is abnormal: Processing method c). 7) Overheating inside the charger: Processing method c). 8) The charger cannot transmit power: Processing method c). 9) Vehicle contactor adhesion: Solution b)

Table C.1 (continued)

Fault level	Fault classification and handling methods
3	Alarm level fault classification and handling methods: 1) Timeout in the charging handshake phase, configuration phase, and charging process timeout Processing method c). 2) Charging ends directly when timeout

Note 1: After the BMS detects a fault, it will, depending on the severity of the fault, choose to provide a charging stop message in the BSM (Battery Status Information) message or the BST (BMS Charging Stop) message, shutting down the charger and entering method b). Alternatively, it can set SPN3090 to SPN3095 in the BSM message to 00 (Battery Status Normal) and SPN3096 to 00 (Charging Disabled), suspending the charger's current output. At this point, the BMS and charger communicate normally until SPN3096 in the BSM message sent by the BMS is 01 (Charging Enabled), at which point the charger's current output is re-enabled. If the waiting time exceeds 10 minutes, the charger suspends charging and saves the reason for the suspension.

Note 2: When the charger detects a charging fault, it immediately sends a CST (Charger Stop Charging) command, shuts down, stops CAN communication, disconnects switches K1, K2, K3, and K4, and then proceeds to the appropriate handling method based on the fault type. Under handling method c), if the charger self-diagnoses and confirms the fault has been resolved, it will re-initiate the handshake identification phase and resume charging. If reconnection fails after three attempts, the operator will be required to check the current status, reconnect the charging connector, and attempt charging again, as per handling method b).

Note 3: When a power outage occurs during charging, even if the power supply automatically recovers after a period of time, manual intervention (processing method b) is required before recharging. Newly charged.

C.3 Untrusted State Handling Methods

When receiving an untrusted state, the receiver maintains the previous state and does not process the data packet.

Appendix D

(Informative Appendix)

Conditions for starting and stopping message transmission

The conditions for starting and stopping sending of various messages are shown in Table D.1.

Table D.1 Message start and stop conditions

Message code	Conditions for starting to send messages	Conditions for stopping sending messages
CHM	low voltage auxiliary power on	Insulation test completed and ready to send CRM
BHM	receives CHM message	Receive CRM message
CRM	insulation verification completed	Receive BCP message
BRM	receives CRM message	Received CRM message with SPN2560=0xAA
BCP	receives a CRM message with SPN2560=0xAA	Receive CML message
BRO	receives CML message	Send BRO message with SPN2829=0xAA and receive CRO message with SPN2830=0xAA
CTS	Receive BCP message	Received BRO message with SPN2829=0xAA
CML	Receive BCP message	Received BRO message with SPN2829=0xAA
CRO	receives BRO message with SPN2829=0xAA	Receive BCL and BCS messages
BCL	Received CRO message with SPN2830=0xAA	Receive CST message (charger automatically stops charging) Or send BST message (BMS automatically stops charging)
BCS	Received CRO message with SPN2830=0xAA	Receive CST message (charger automatically stops charging) Or send BST message (BMS automatically stops charging)
CCS	receives BCL and BCS messages	Receive BST message (BMS automatically stops charging) Or send CST message (charger automatically stops charging)
BSM	Receive CCS message	Receive CST message (charger automatically stops charging) Or send BST message (BMS automatically stops charging)
BMV	Receive CCS message	Receive CST message (charger automatically stops charging) Or send BST message (BMS automatically stops charging)
BMT	Receive CCS message	Receive CST message (charger automatically stops charging) Or send BST message (BMS automatically stops charging)
BSP	Receive CCS message	Receive CST message (charger automatically stops charging) Or send BST message (BMS automatically stops charging)
BST	When BMS needs to stop charging (BMS actively stops charging) or receives CST (charger actively stops charging)	Receiving CST message (BMS automatically stops charging) or sending BSD message (charger automatically stops charging)
CST	When the charger needs to stop charging (charger actively stops charging); or when BST is received (BMS actively stops charging)	Receive BSD message
BSD	receives CST message	1) BMS receives the charger identification message sent by the charger (CRM) 2) Or the auxiliary power output cannot be detected
CSD	receives BSD message	1) Restart handshake and send CRM frame 2) Or turn off the auxiliary power

Table D.1 (continued)

Message code	Conditions for starting to send messages	Conditions for stopping sending messages
BEM	When BMS detects an error in the message	1) BMS receives the charger identification message sent by the charger (CRM) 2) Or the auxiliary power output cannot be detected
CEM	When the charger detects an error in the message	1) Restart handshake and send CRM frame 2) Or turn off the auxiliary power supply